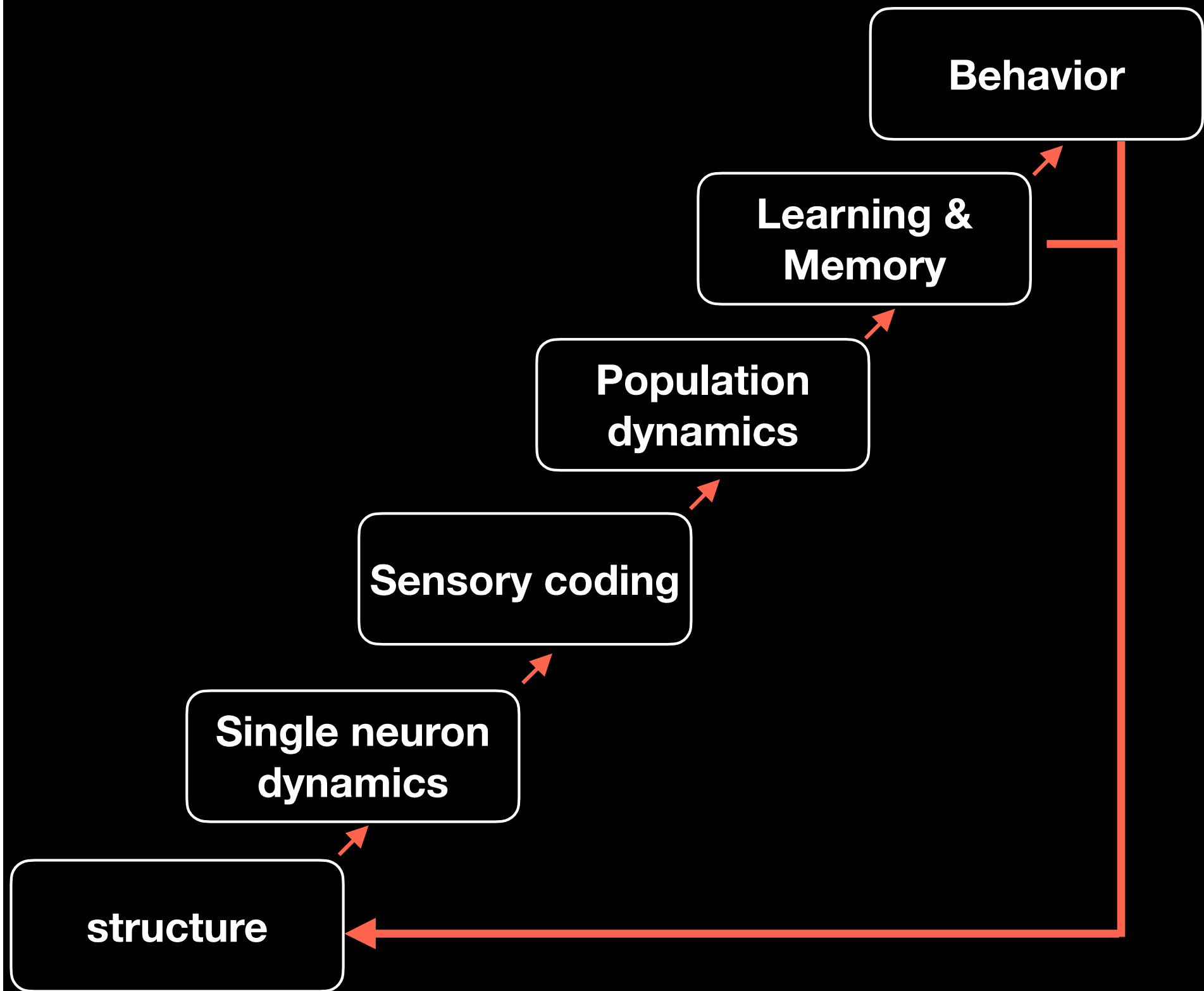




Hierarchical Behavior Control

Quan Wen

A prescient comment

Hierarchical organization, a candidate principle for ethology

Richard Dawkins

(1977)

If we look far into the future of our science, what will it mean to say we “understand” the mechanism of behaviour? **The obvious answer is what may be called the neurophysiologist’s nirvana: the complete wiring diagram of the nervous system of a species**, every synapse labelled as excitatory or inhibitory; presumably also a graph, for each axon, of nerve impulses as a function of time during the course of each behaviour pattern. ... and because we are still in the early stages, the ultimate conclusion does not worry us. **But it would not constitute understanding of how behaviour works in any real sense at all. No man could hold such a mass of detail in his head. Real understanding will only come from distillation of general principles at a higher level**, to parallel for example the great principles of genetics ...

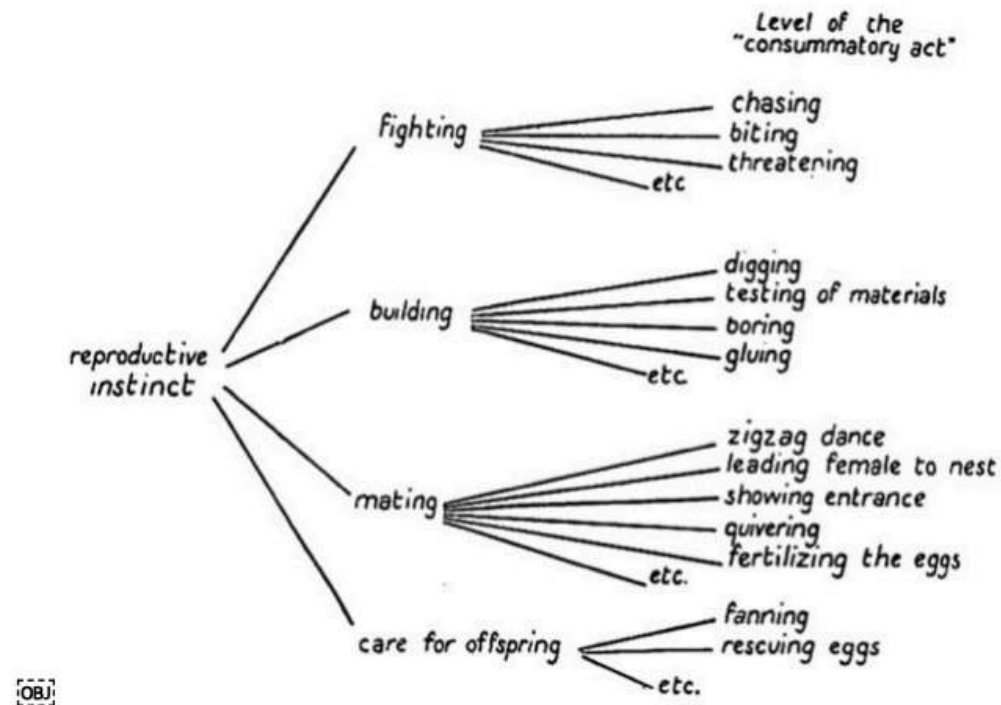


Fig. 89. The principle of hierarchical organization illustrated by the reproductive instinct of the male three-spined stickleback. After Tinbergen, 1942.

The study of instinct, Tinbergen

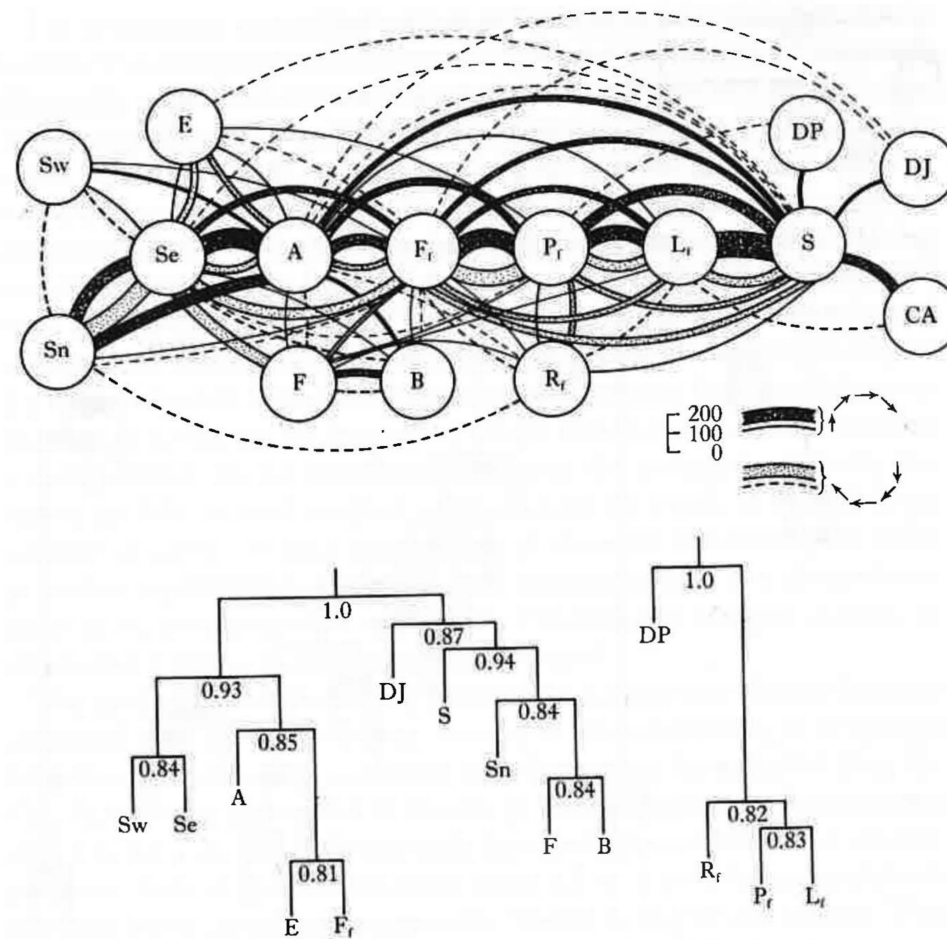
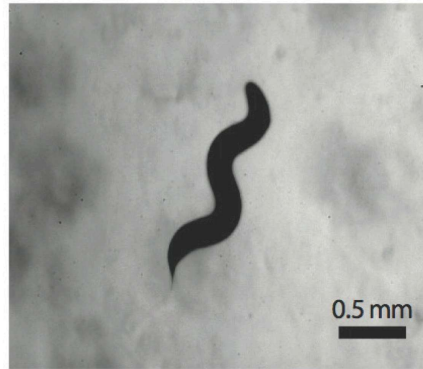


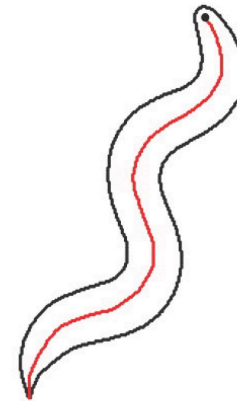
Fig. 7. (b) Mutual Replaceability Cluster Analysis on data from Baerends, Brouwer & Waterbolk (1955), on courtship behaviour of male guppies, *Lebistes reticulatus*. As Fig. 7a, except Spearman rank correlations used. Upper part of figure: representation of transition frequencies by original authors. Behaviour symbols are those of original authors.

How to quantitatively characterize behaviors?

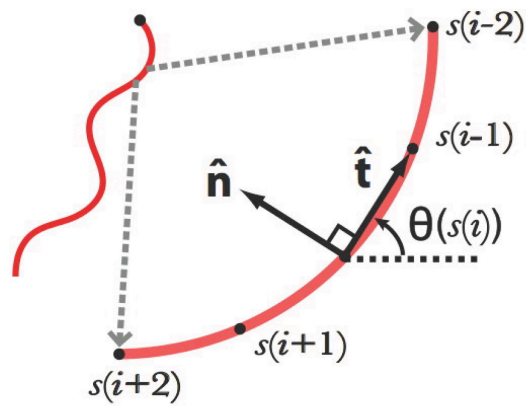
The eigenworm



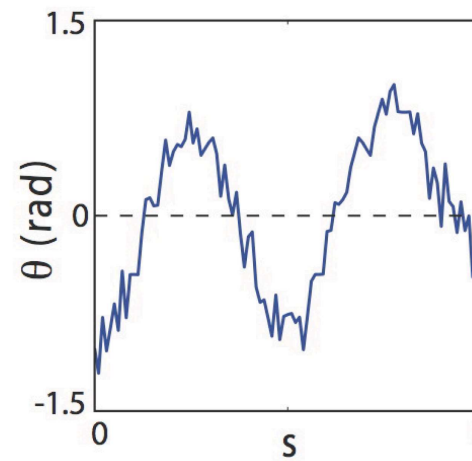
A



B

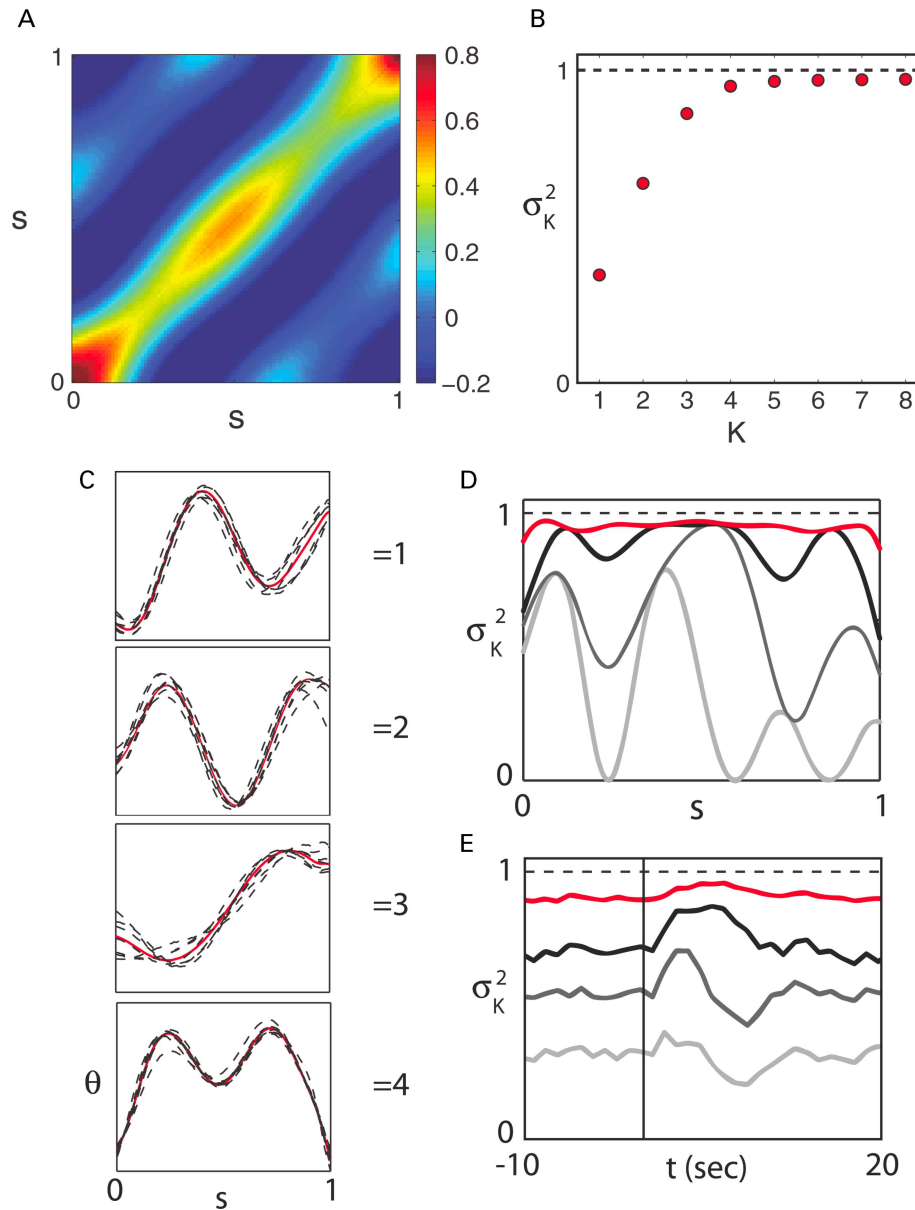


C



D

The eigenworm

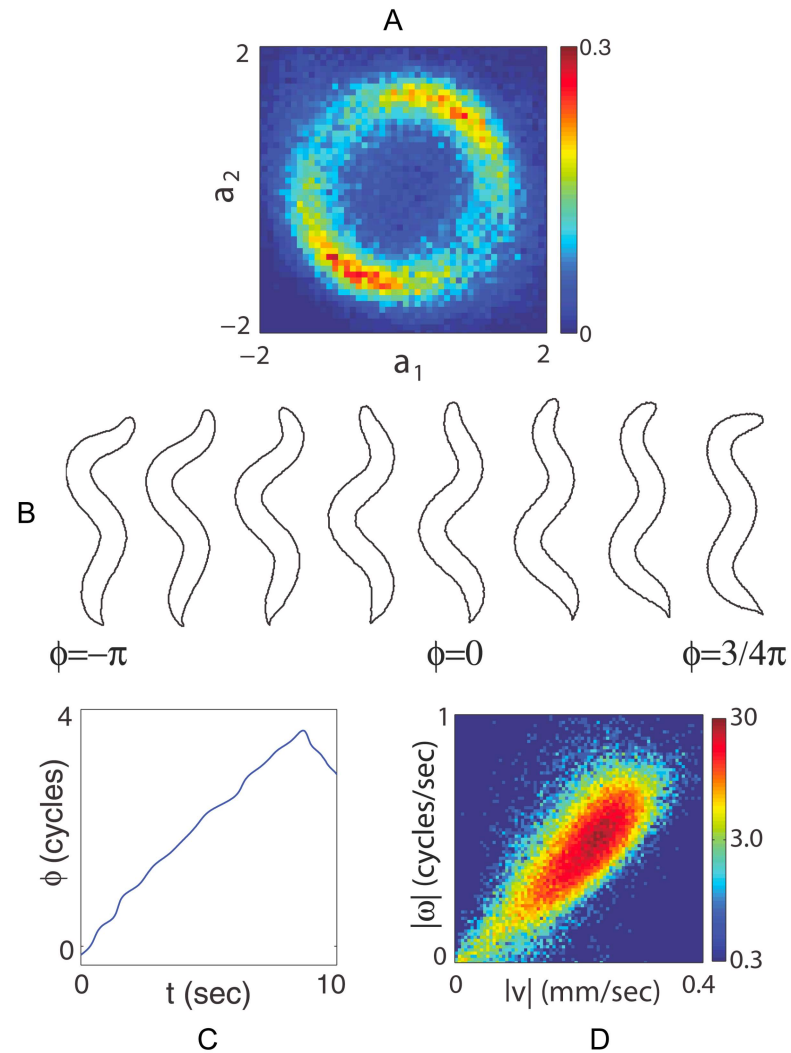


$$C_{ij} = \langle (\theta_i - \bar{\theta})(\theta_j - \bar{\theta}) \rangle$$

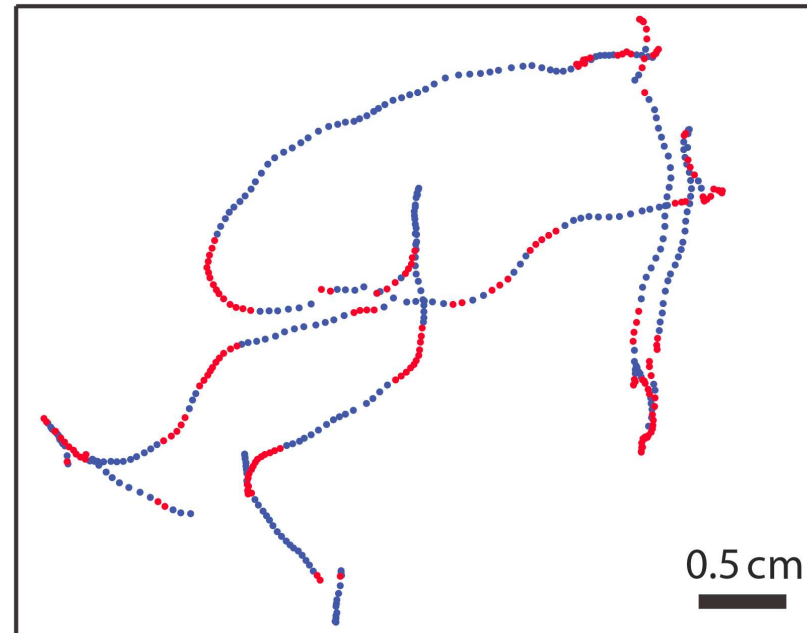
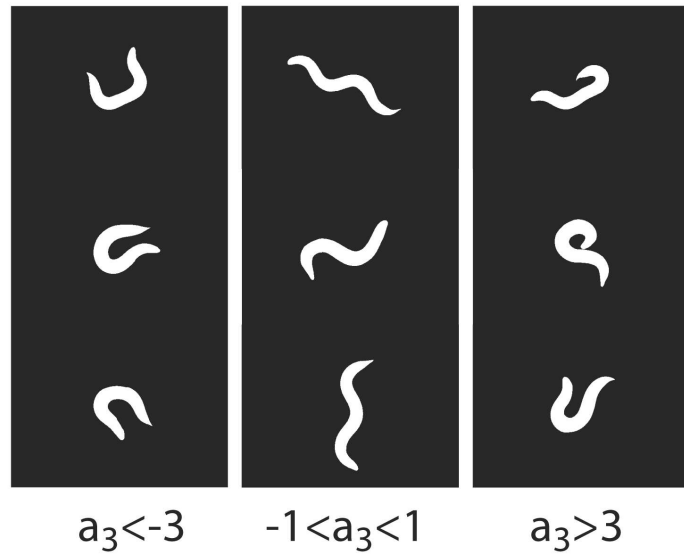
$$C = \sum_{i=1}^N \lambda_i \Theta_i \Theta_i^T$$

$$\vec{\theta}(t) \approx \sum_{i=1}^D a_i(t) \Theta_i,$$

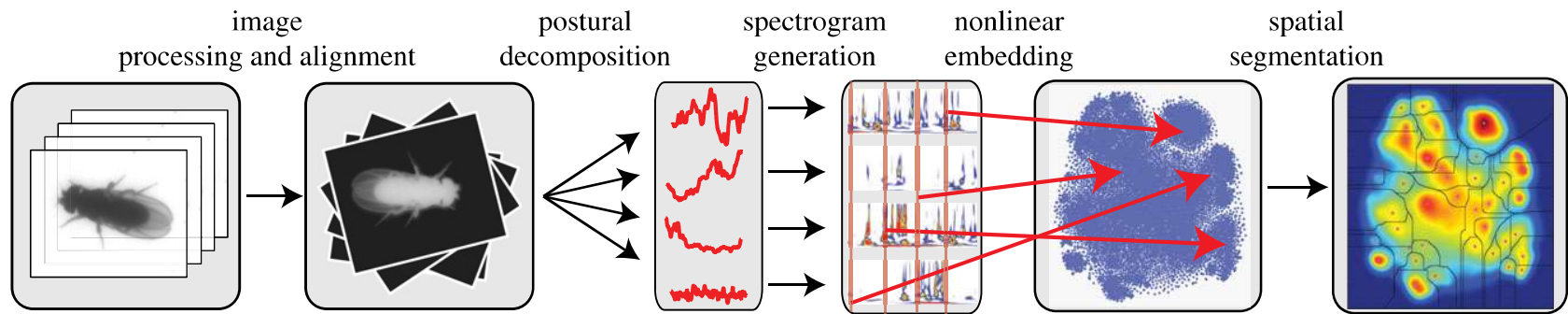
What does each eigenmode mean?



What does each eigenmode mean?

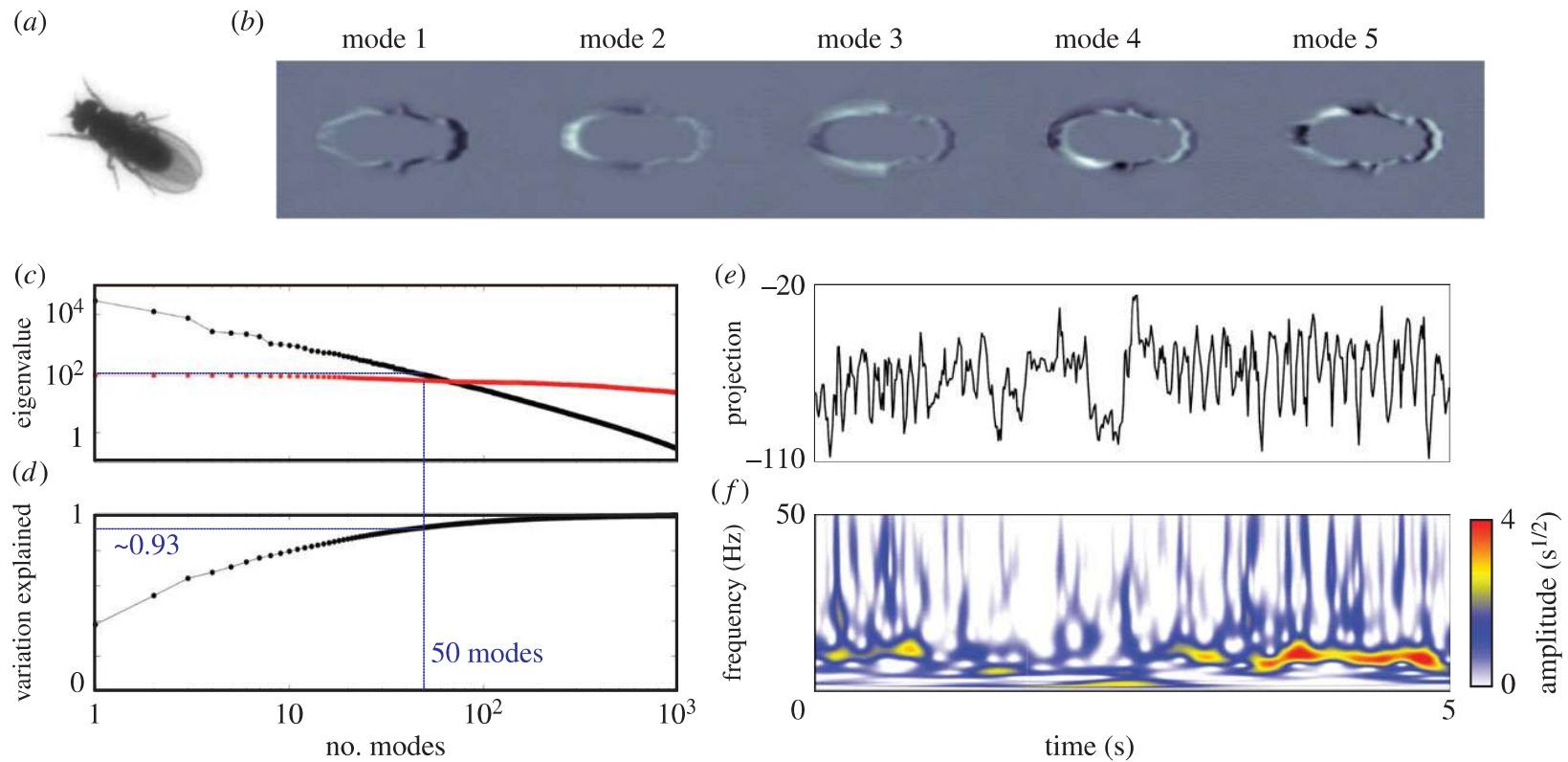


how to categorize behaviors?

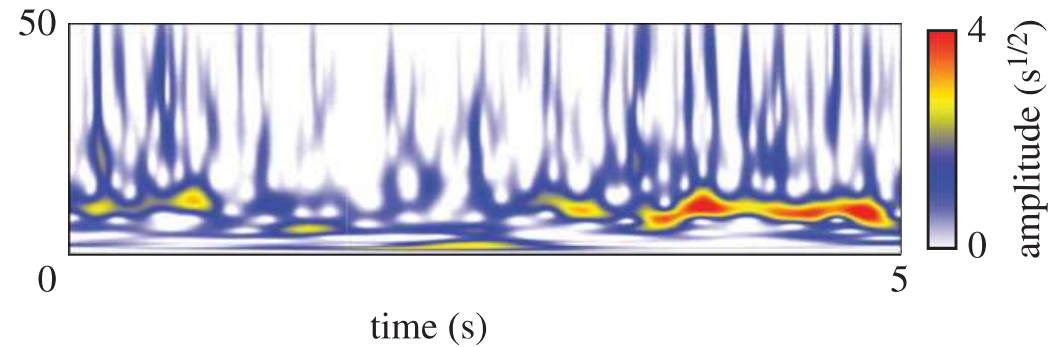


From crawling worm to walking fly

1. Extract spatiotemporal features using PCA



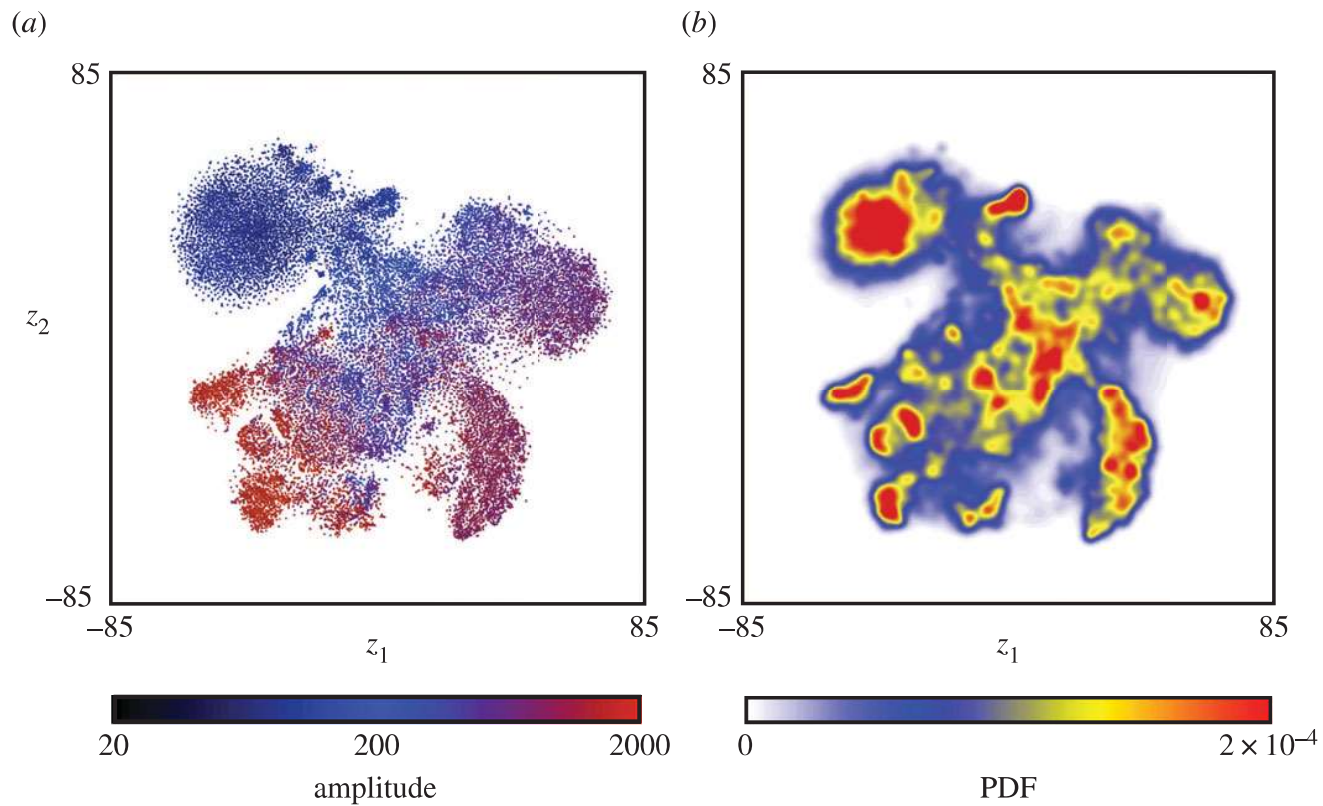
2. Characterize a pairwise distance function



$$\hat{S}(k, f, t) = \frac{S(k, f, t)}{\sum_{k,f} S(k, f, t)}$$

$$d(t_1, t_2) = D_{KL}(t_2 || t_1) = \sum_{k,f} \hat{S}(k, f, t_1) \log \frac{\hat{S}(k, f, t_1)}{\hat{S}(k, f, t_2)}$$

3. t-SNE embedding for visualization



t-distributed Stochastic neighbor embedding

tSNE (van der Maaten & Hinton, 2008)

t-distribution with 1 degree of freedom
= Cauchy distribution



$$p_{i|j} = \frac{\exp[-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i]}{\sum_{k \neq i} \exp[-\|\mathbf{x}_i - \mathbf{x}_k\|^2 / 2\sigma_i]}$$

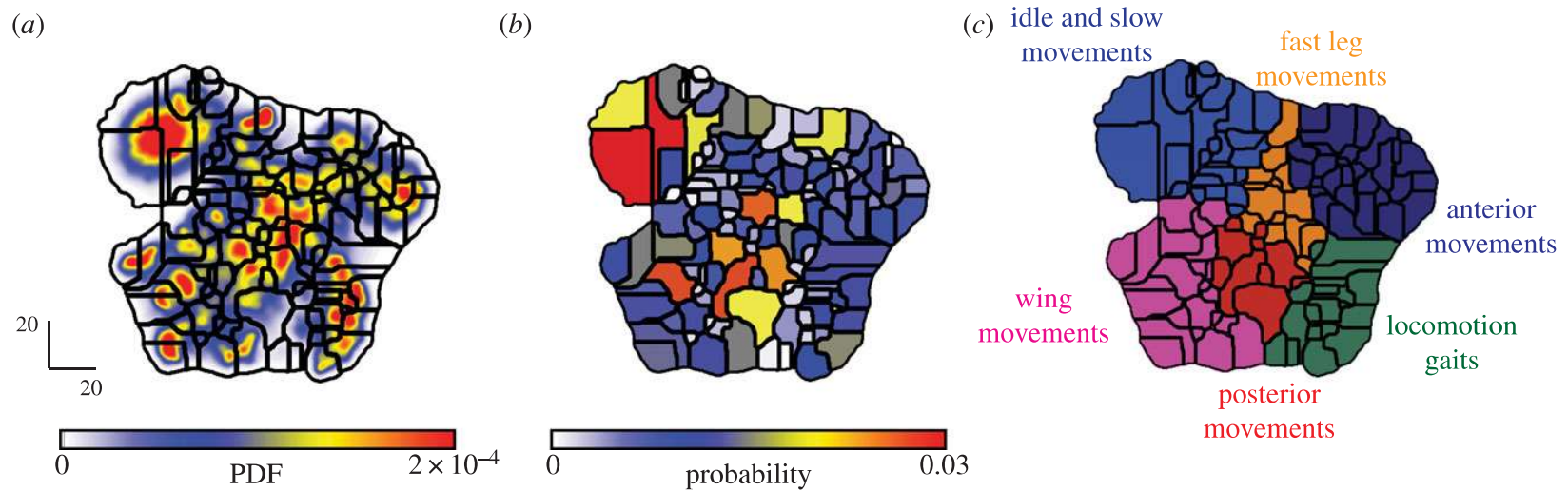
$$q_{i|j} = \frac{\left(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2\right)^{-1}}{\sum_{k \neq i} \left(1 + \|\mathbf{y}_i - \mathbf{y}_k\|^2\right)^{-1}}$$

learn $\mathbf{y}_i$

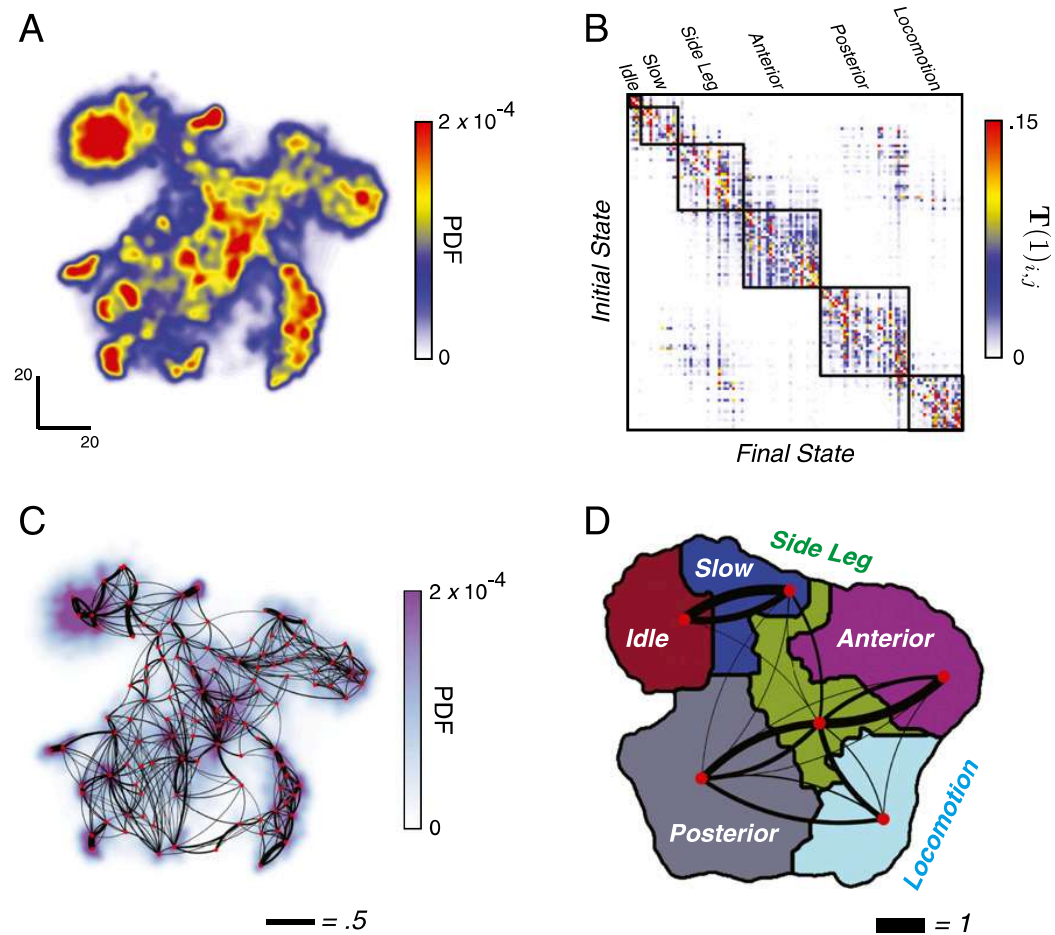
$$D(\mathbf{P} \parallel \mathbf{Q}) = \sum_i \sum_j \log p_{i|j} \frac{p_{i|j}}{q_{i|j}}$$

Kullback-Leibler
divergence

4. Segmentation on the t-SNE space



Transitions between different states



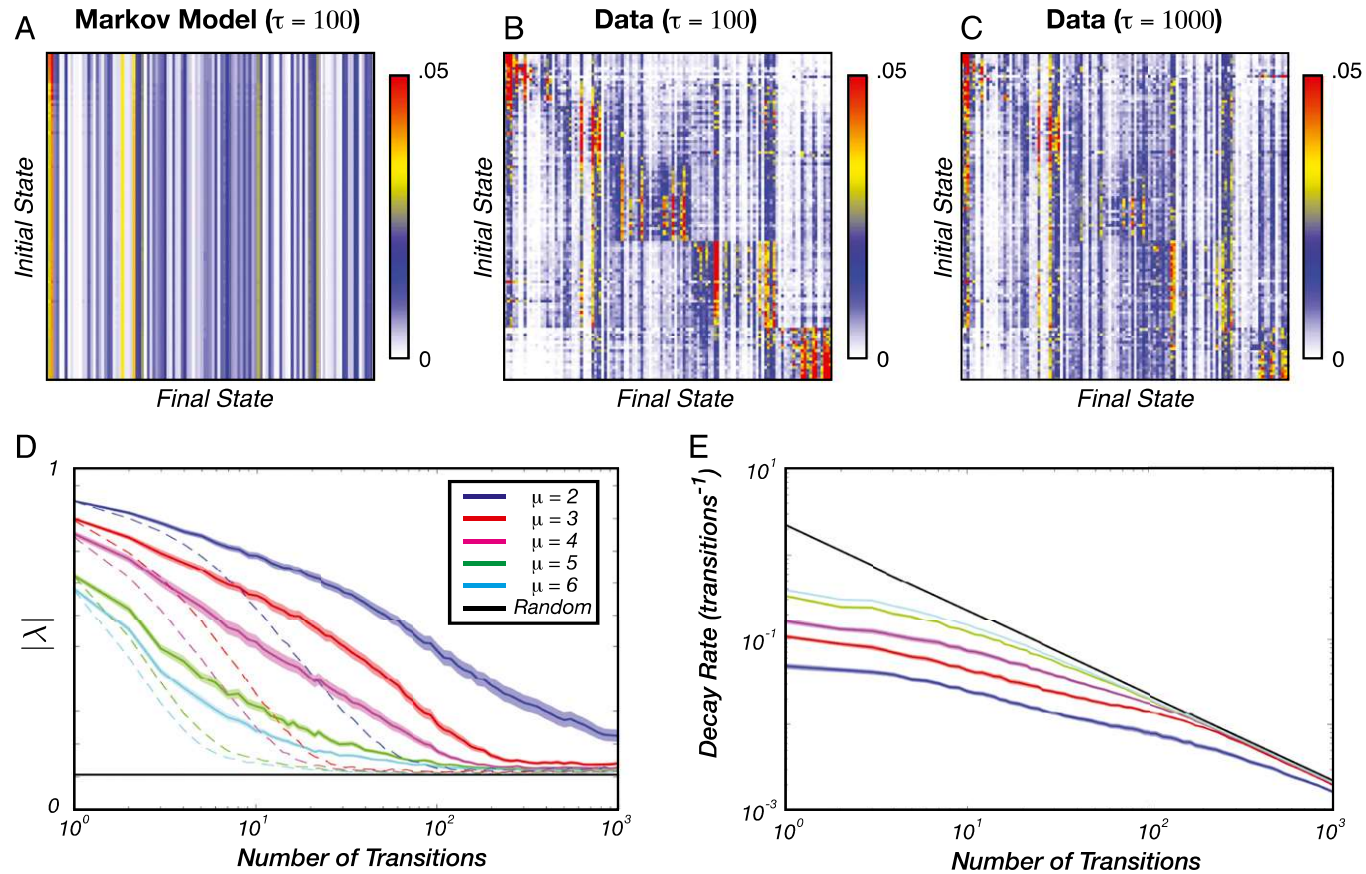
Stochastic matrix

$$P = \begin{bmatrix} P_{1,1} & P_{1,2} & \dots & P_{1,j} & \dots & P_{1,\alpha} \\ P_{2,1} & P_{2,2} & \dots & P_{2,j} & \dots & P_{2,\alpha} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ P_{i,1} & P_{i,2} & \dots & P_{i,j} & \dots & P_{i,\alpha} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ P_{\alpha,1} & P_{\alpha,2} & \dots & P_{\alpha,j} & \dots & P_{\alpha,\alpha} \end{bmatrix}, \quad \sum_{j=1}^{\alpha} P_{i,j} = 1;$$

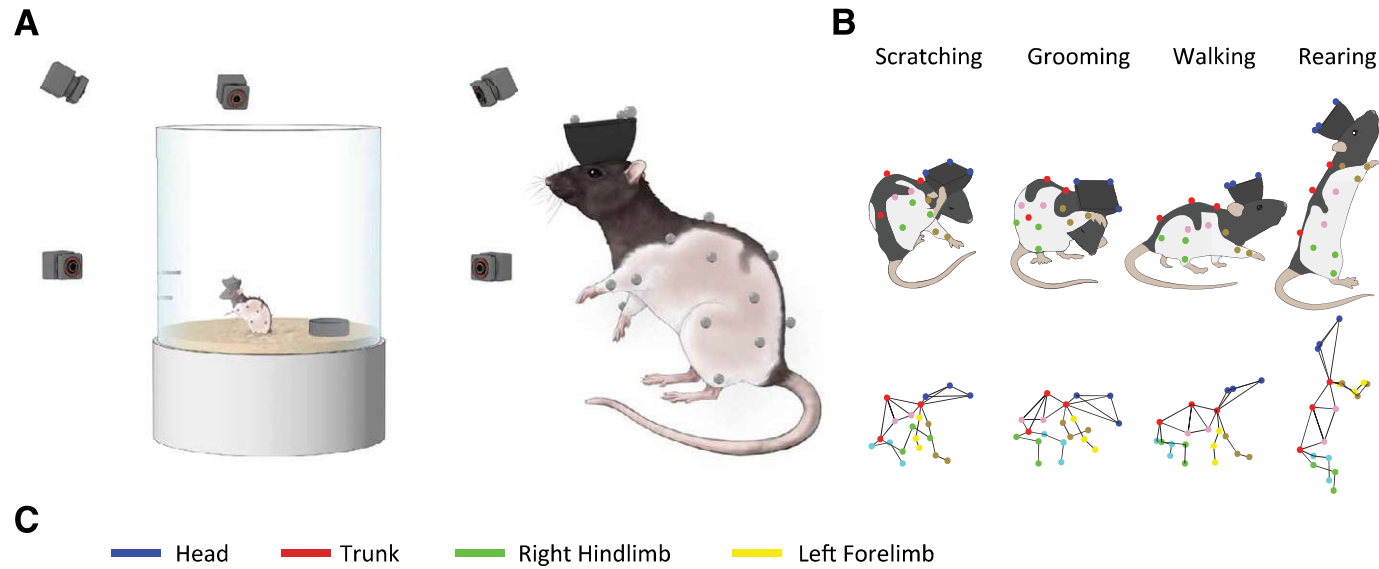
$$P = \sum_{i=1} \lambda_i \vec{v}^i \vec{\pi}^i$$

$$P_{ij}^2 = \sum_k P_{ik} P_{kj}$$

Non-Markovian dynamics

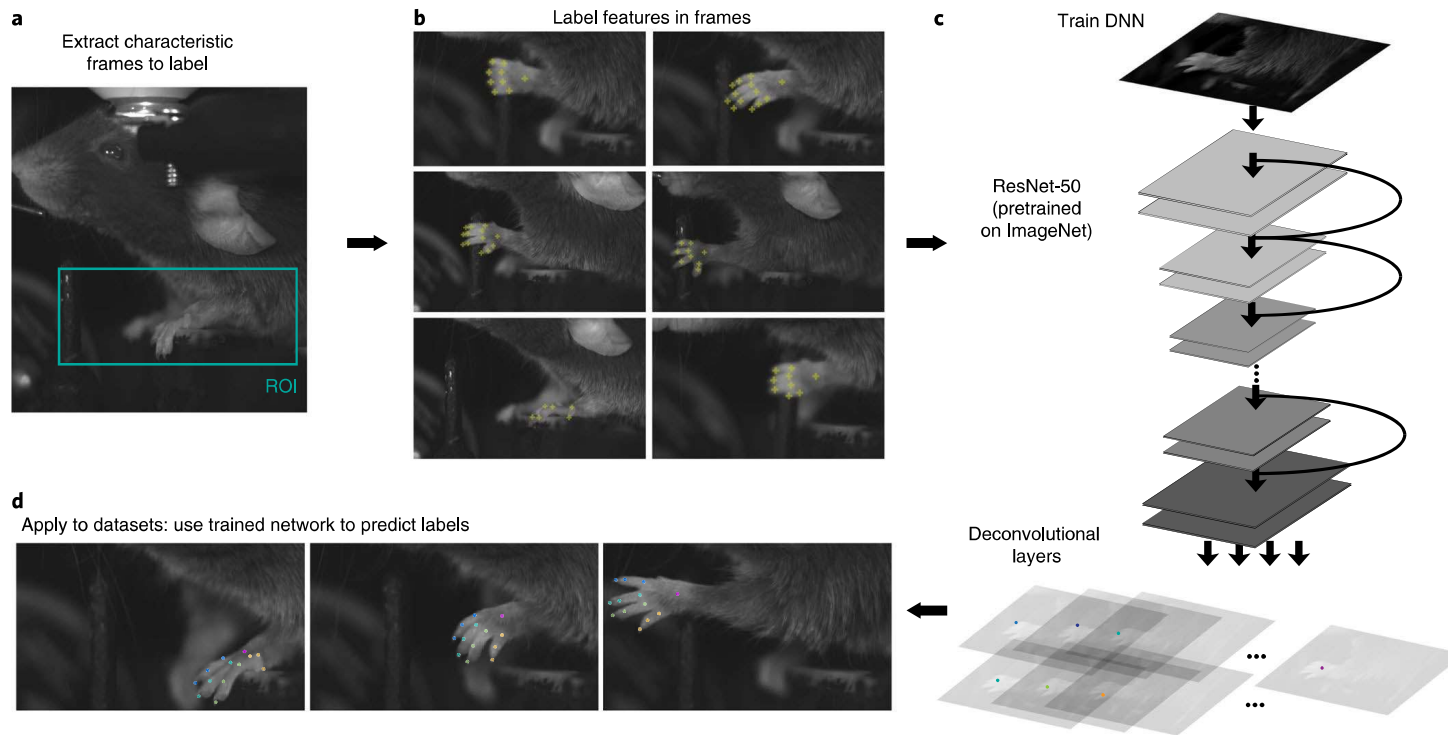


From fly to 3D behavior of mice



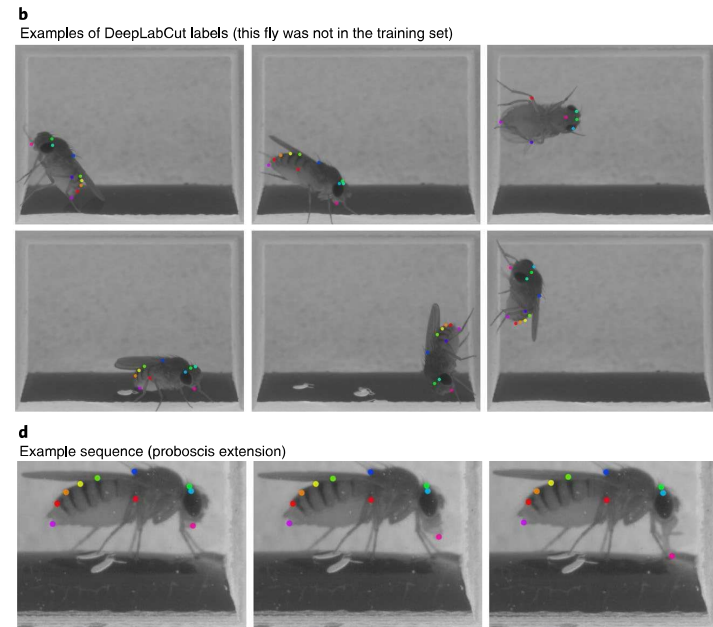
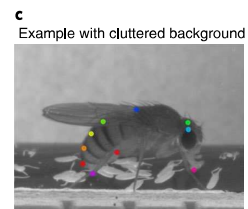
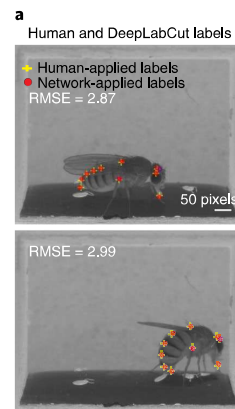
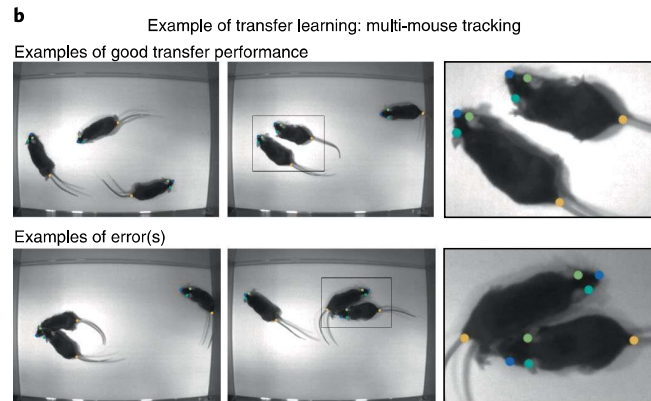
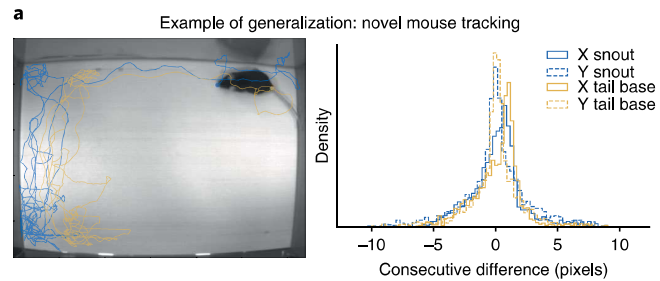
DeepLabCut: markerless pose estimation of user-defined body parts

DeepLabCut: markerless tracking toolbox

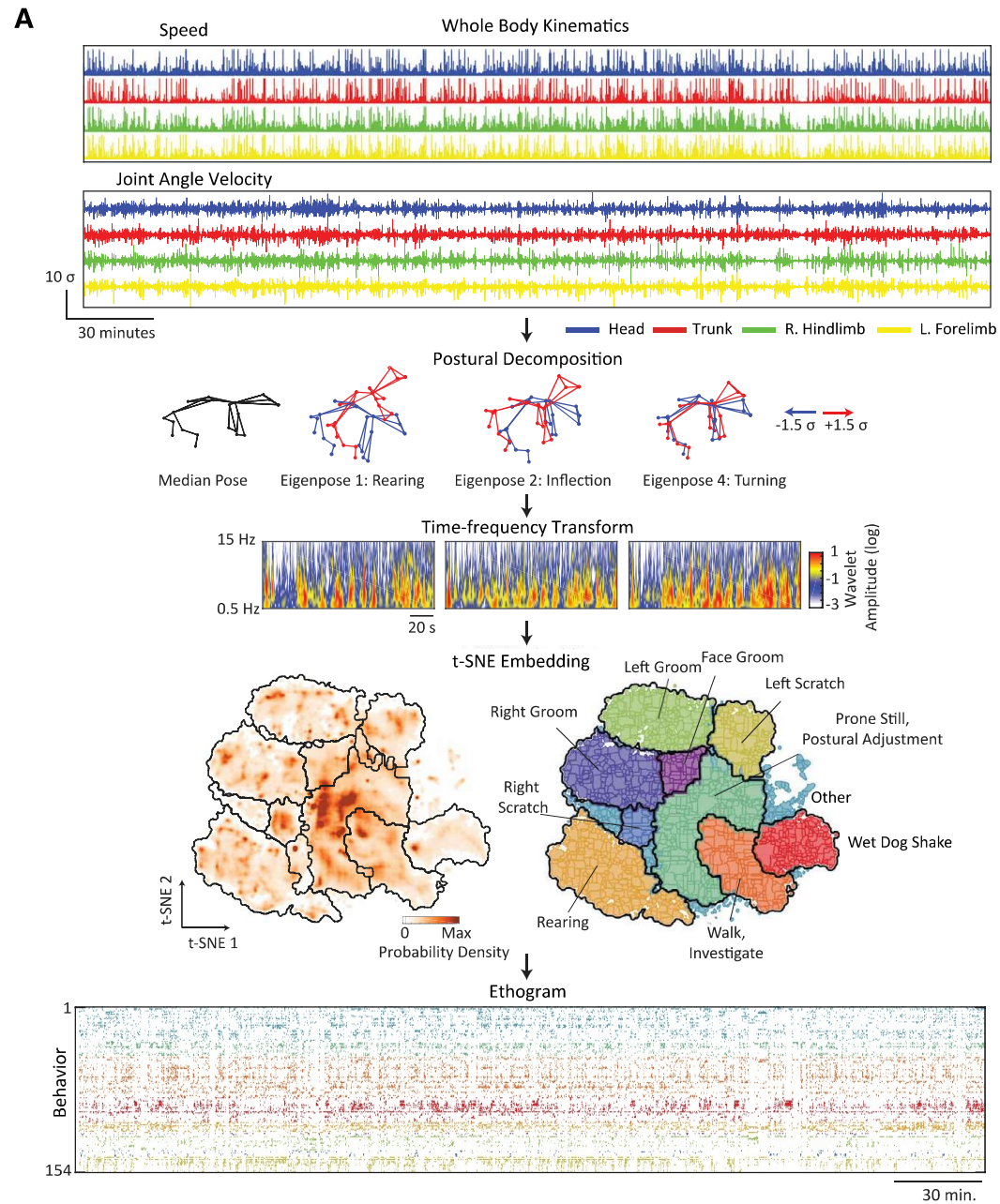


Mathis, et. al., 2018

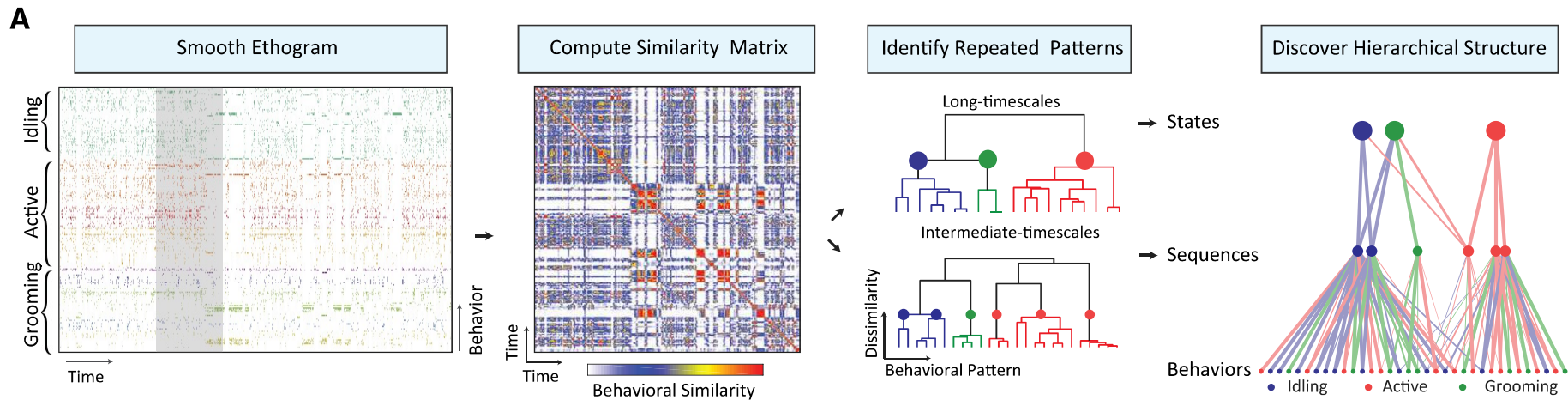
DeepLabCut: markerless pose estimation of user-defined body parts



Posture analysis

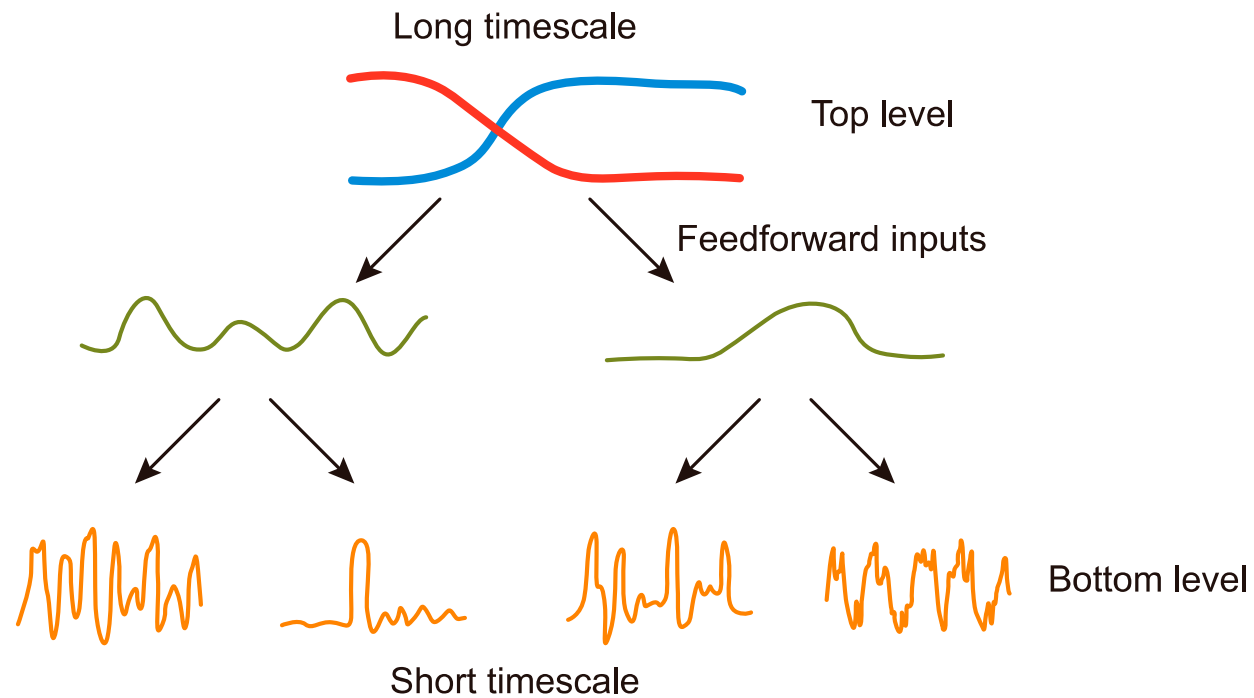


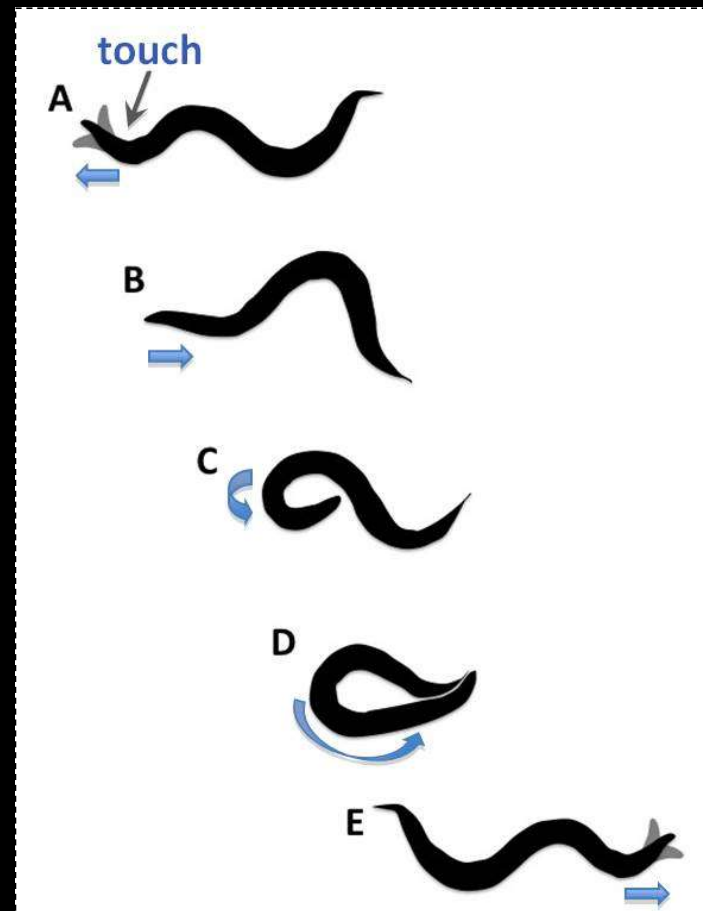
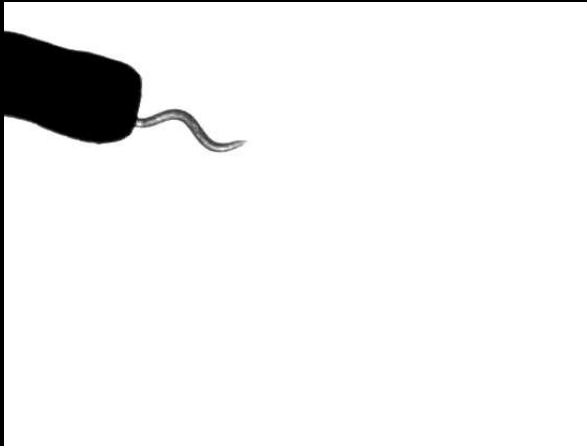
Similarity between different behaviors



Neural representation of movements

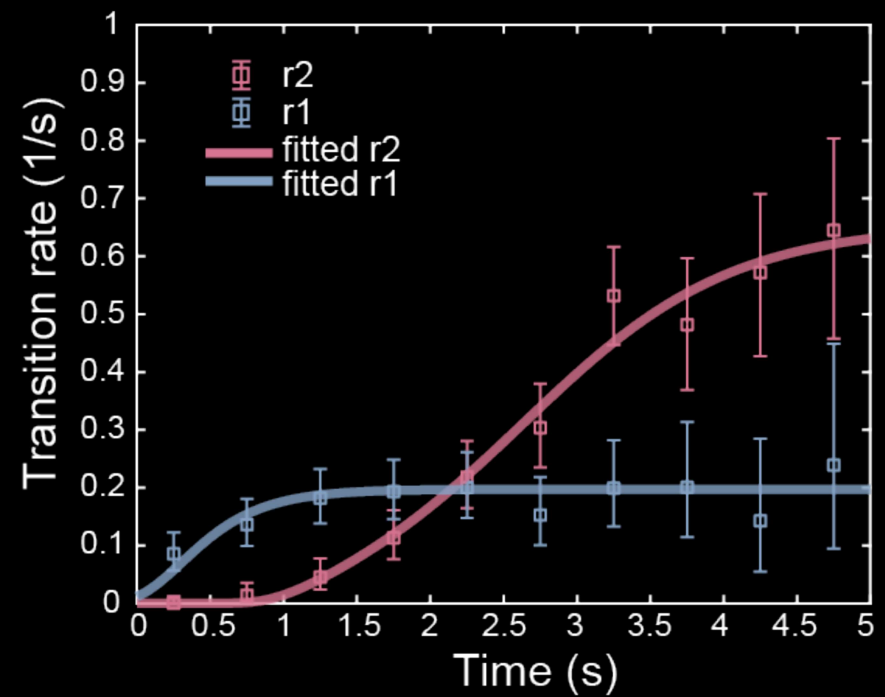
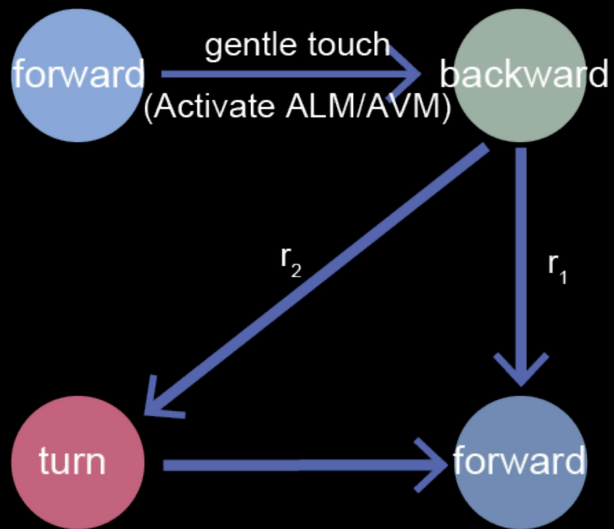
The hierarchical architecture for movement control



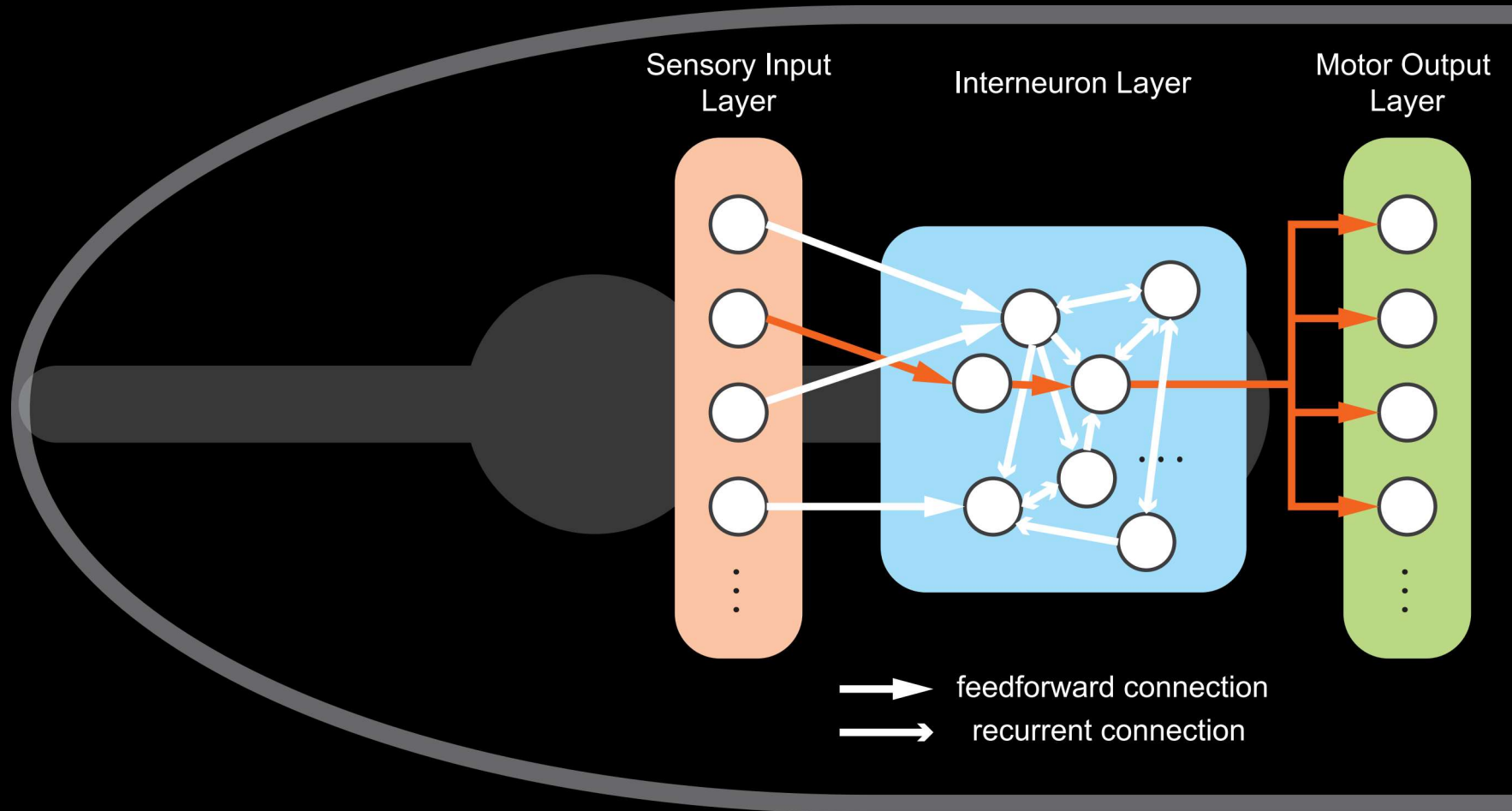


Motor Sequence at a Long Timescale

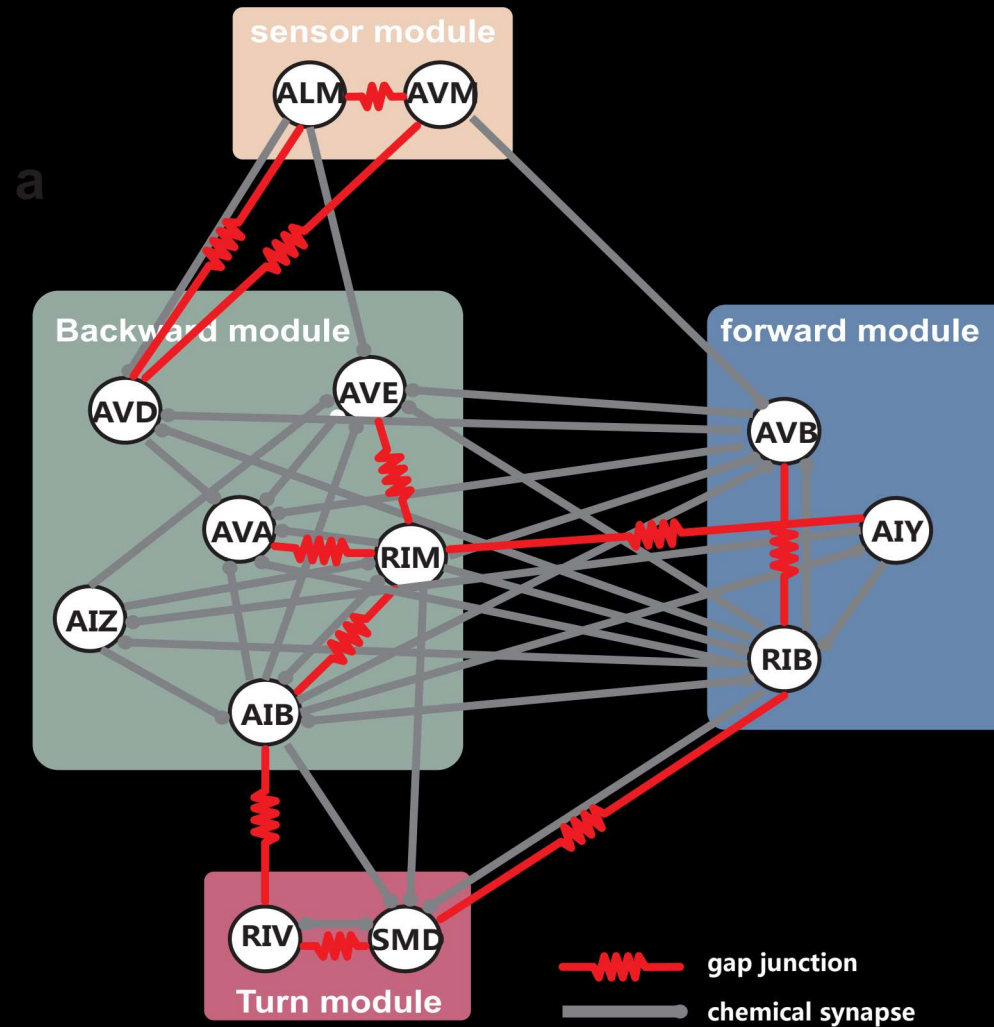
Type I and Type II transitions



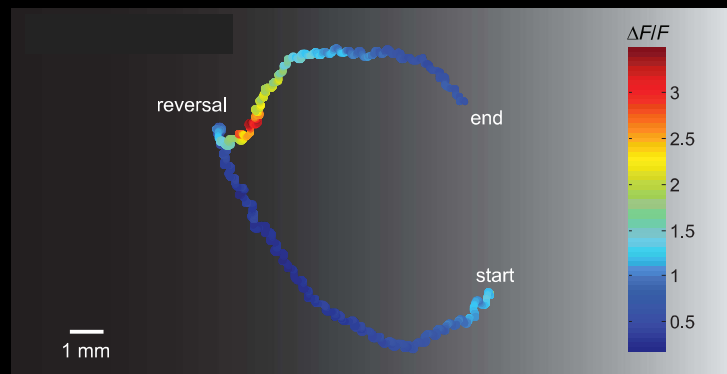
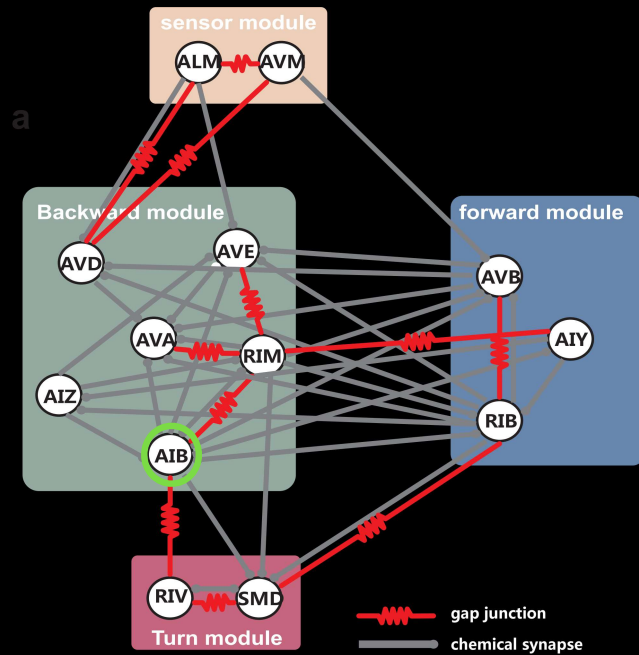
How does a neural circuit perform action selection and motor state transition?



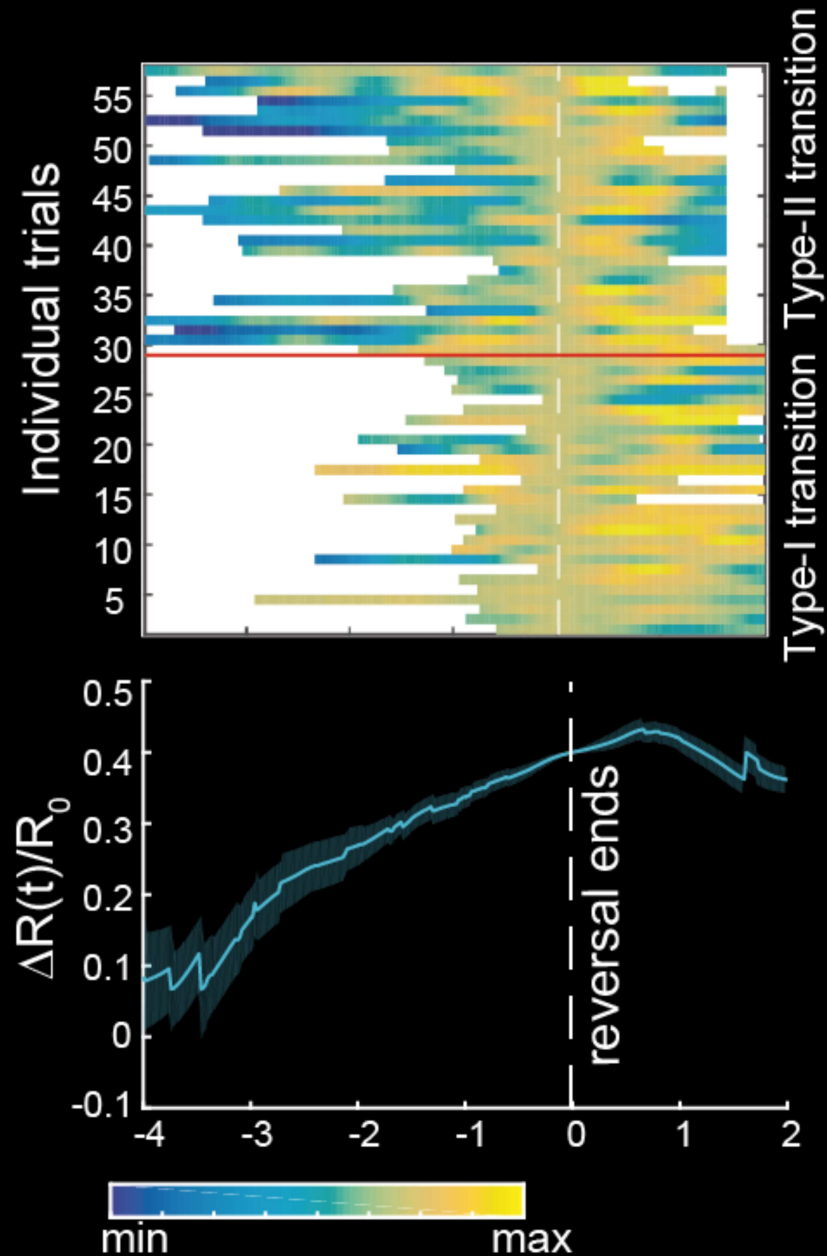
Rapid segregation from sensory to motor representation



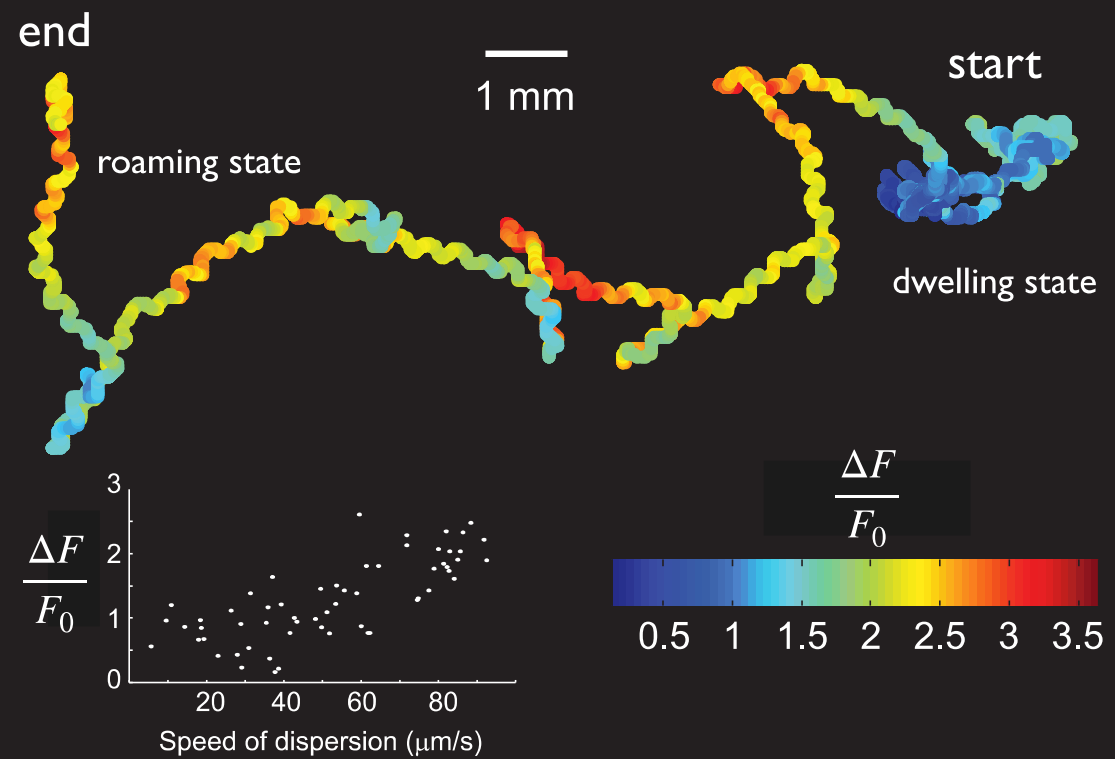
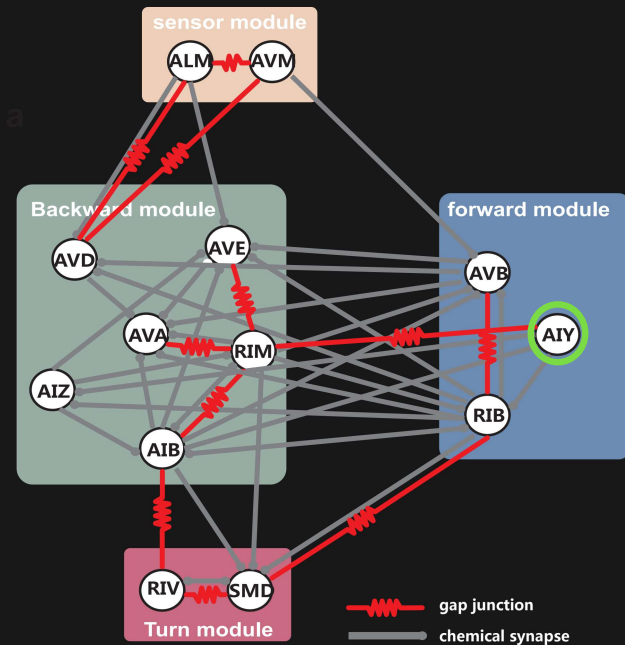
Gray JM et. al., PNAS, 2005
Kawano T et. al., Neuron 2011
Luo LJ et. al., Neuron, 2014
Kato S et. al., Cell, 2015
Li ZY et. al., Cell, 2014
Xu et. al., PNAS, 2018
Wang et. al., eLife 2020

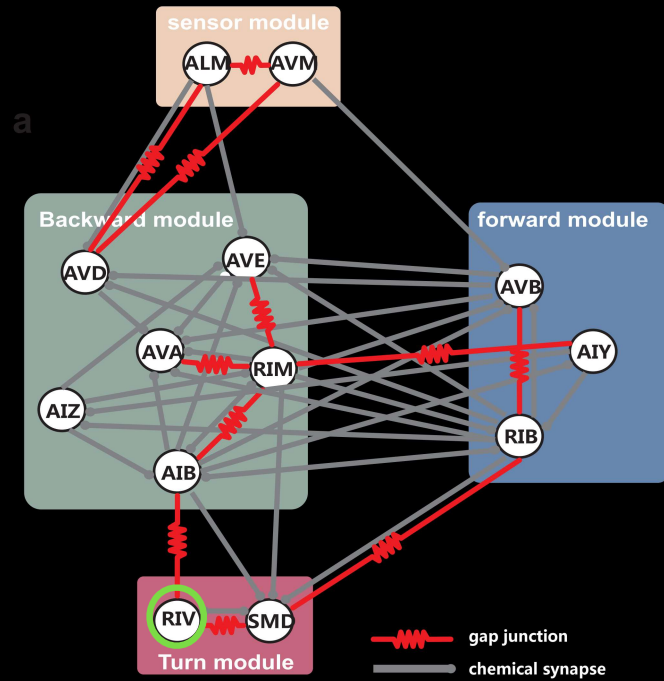


Some examples

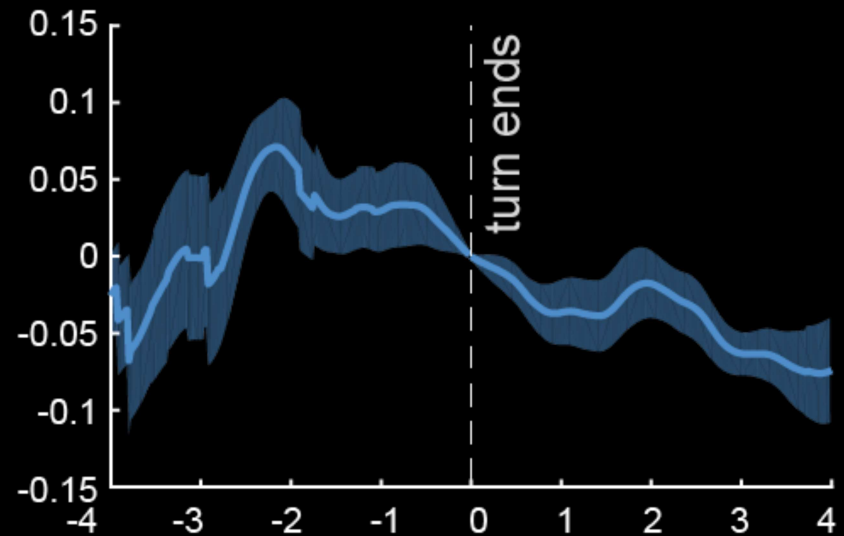
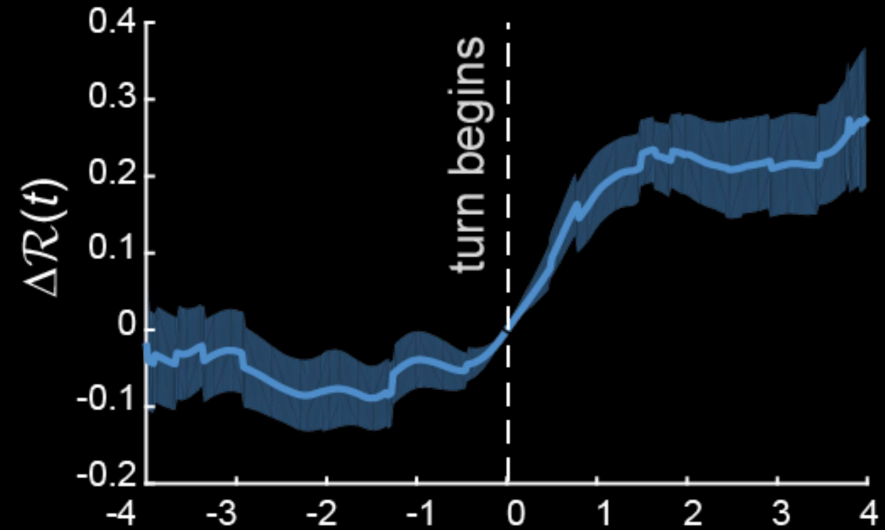


Some examples





Some examples



AVB descending inputs provide command signals

Recording



Psra-1::Chrimson

32233

Activation

Recording

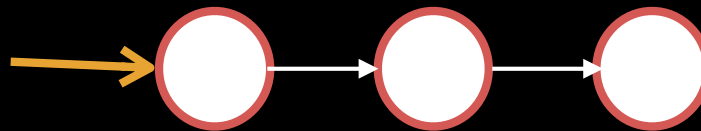
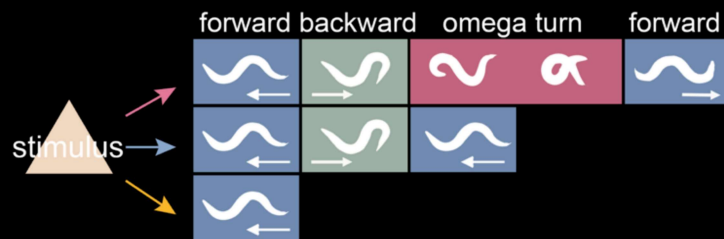


Psra-1::GtACR2

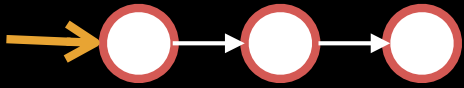
14566

Inhibition

The synaptic chain model



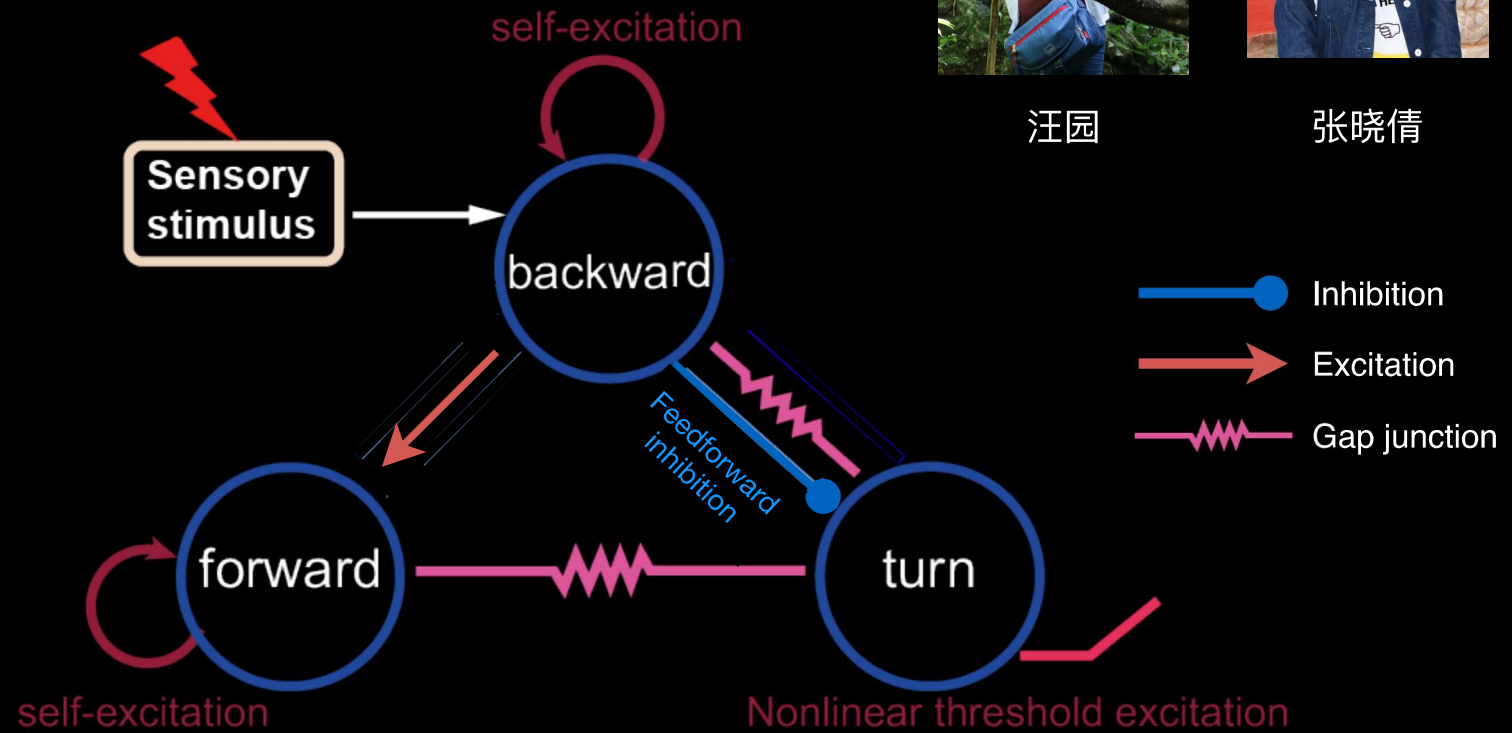
Abele M, 1991
*Corticonics: Neural Circuit
of the Cerebral Cortex*
Cambridge University Press



汪园



张晓倩



Wang et. al., eLife 2020
Meng et. al., bioRxiv, 2022

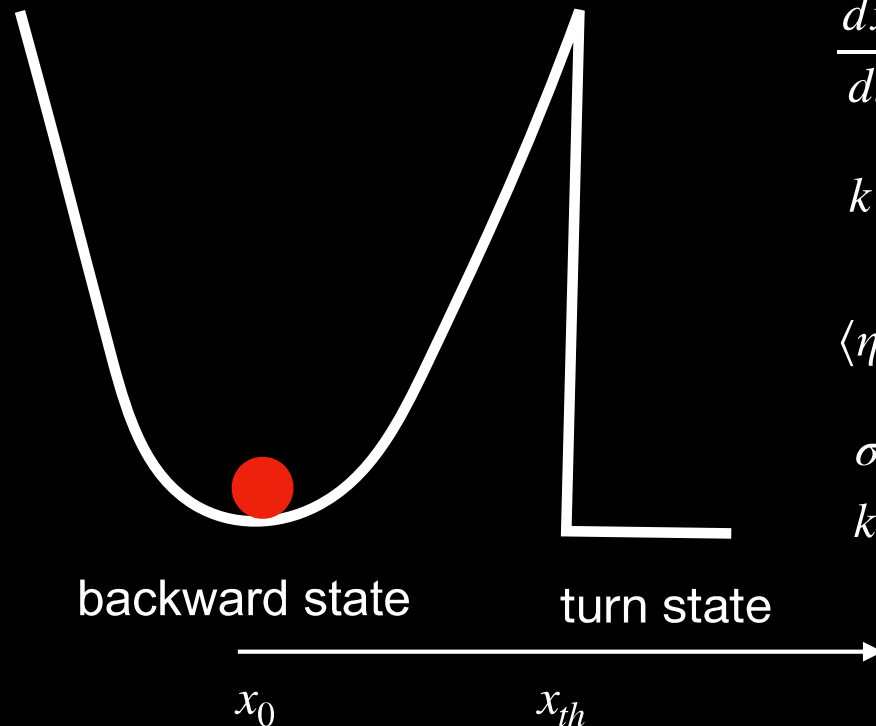
Energy landscape and type-II transition



辛琦

$$C_m \frac{dx}{dt} = - \overset{\text{gap junction inputs}}{G_g(x - V^{AIB})} - \overset{\text{inhibitory inputs}}{G_i(x - E^i)} - G_l(x - E^l) + I_{noise}, \quad x < x_{th}$$

x : membrane potential of the neuron in the turning module, say RIV



$$\frac{dx}{dt} = -k(x - x_0) + \eta; \text{ Langevin equation}$$

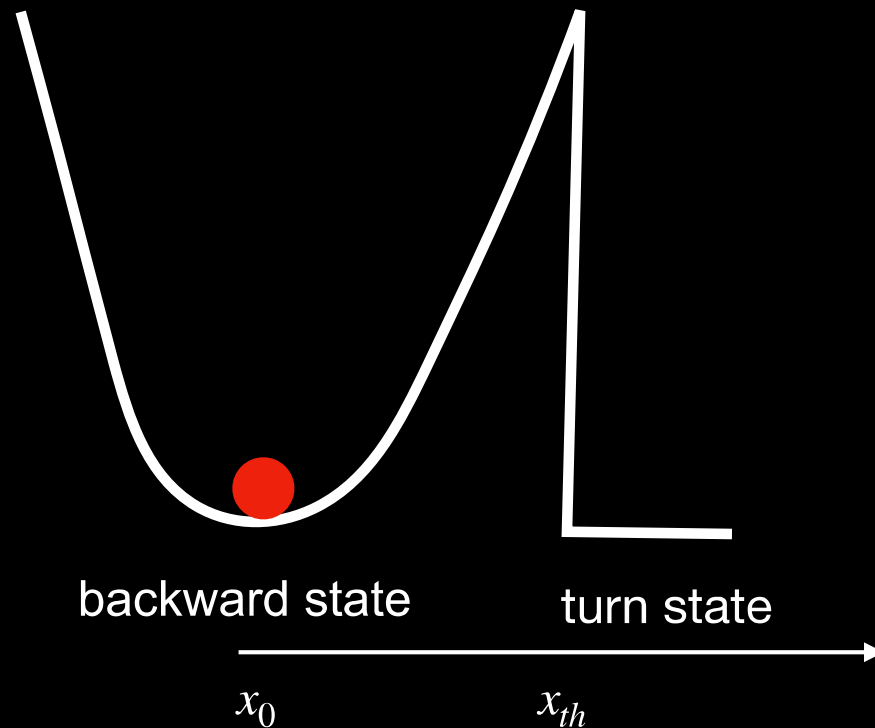
$$k = \frac{1}{C_m} \sum_n G_n$$

$$\langle \eta(t) \eta(t') \rangle = 2\sigma^2 \delta(t - t')$$

σ : noise amplitude

k : steepness of the well

Energy landscape and type-II transition



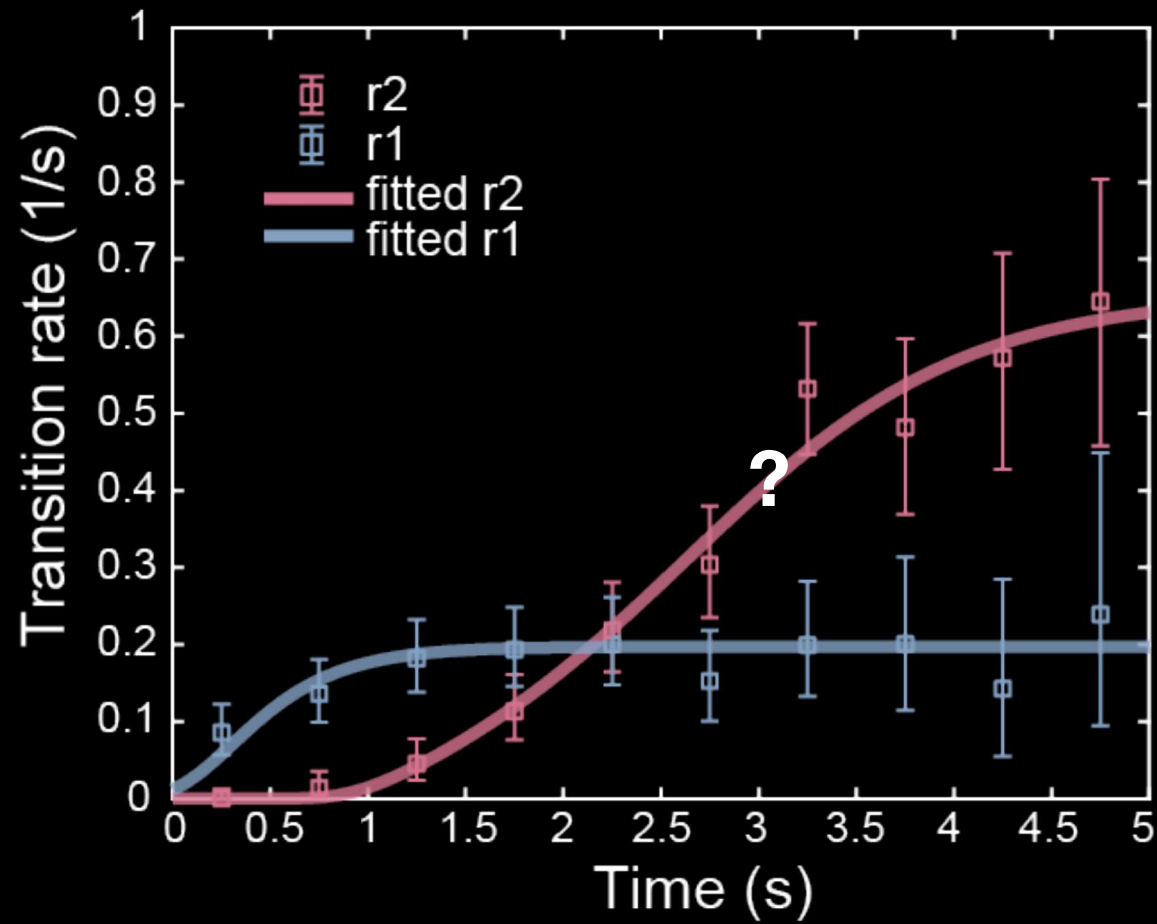
$$r_2 \approx \frac{k}{\pi} \operatorname{erfi}^{-1} \left[\sqrt{\frac{k}{2\sigma^2}} (x_{th} - x_0) \right]$$

σ : **noise amplitude**

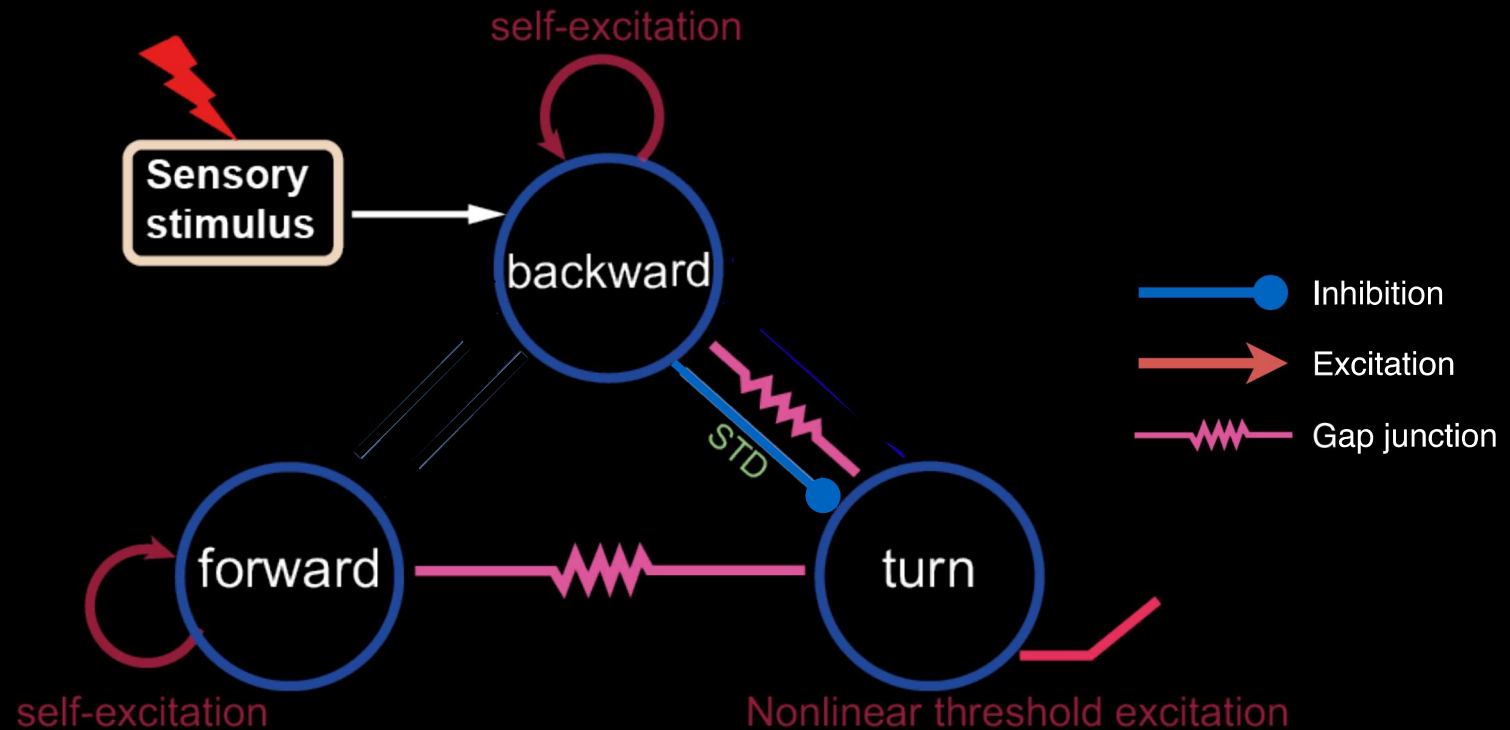
k : **steepness of the well**

$$\operatorname{erfi}(x) = \int_0^x e^{z^2} dz$$

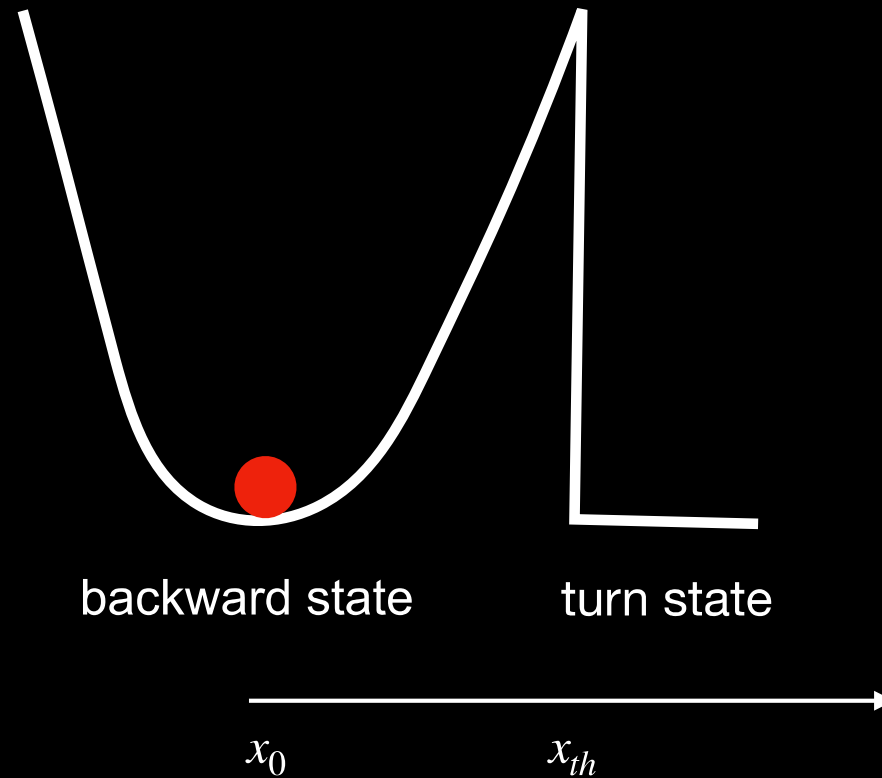
The rising phase of type-II transition ?



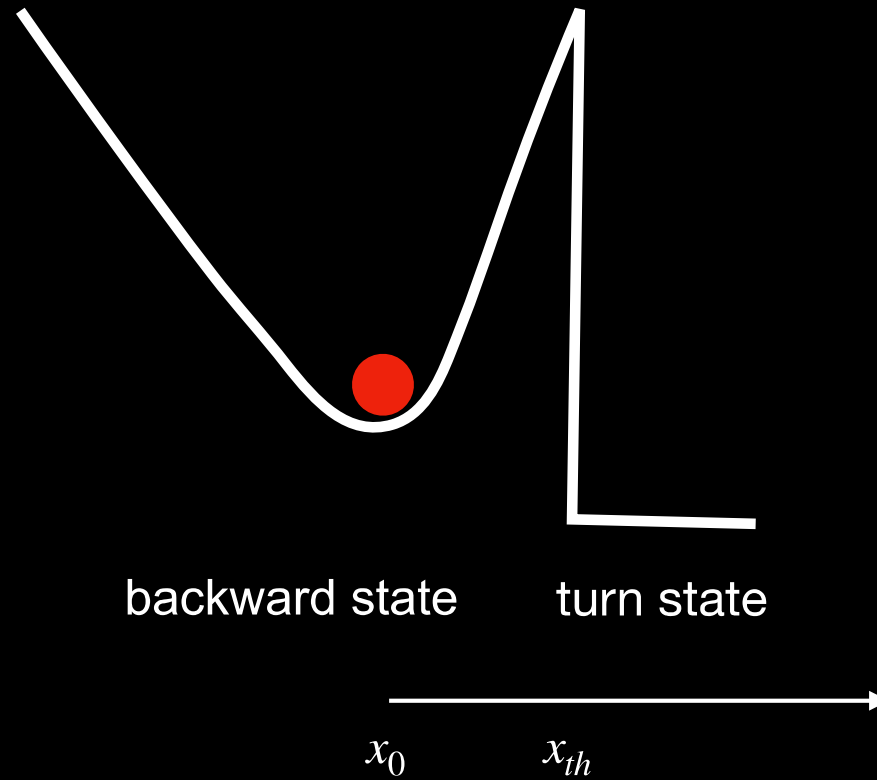
Introducing short term depression (STD) in feedforward inhibition



STD allows energy landscape to change over time

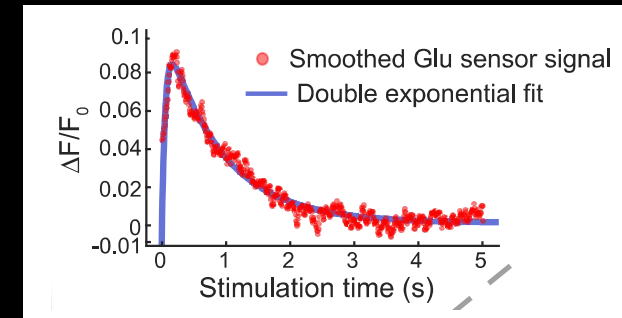
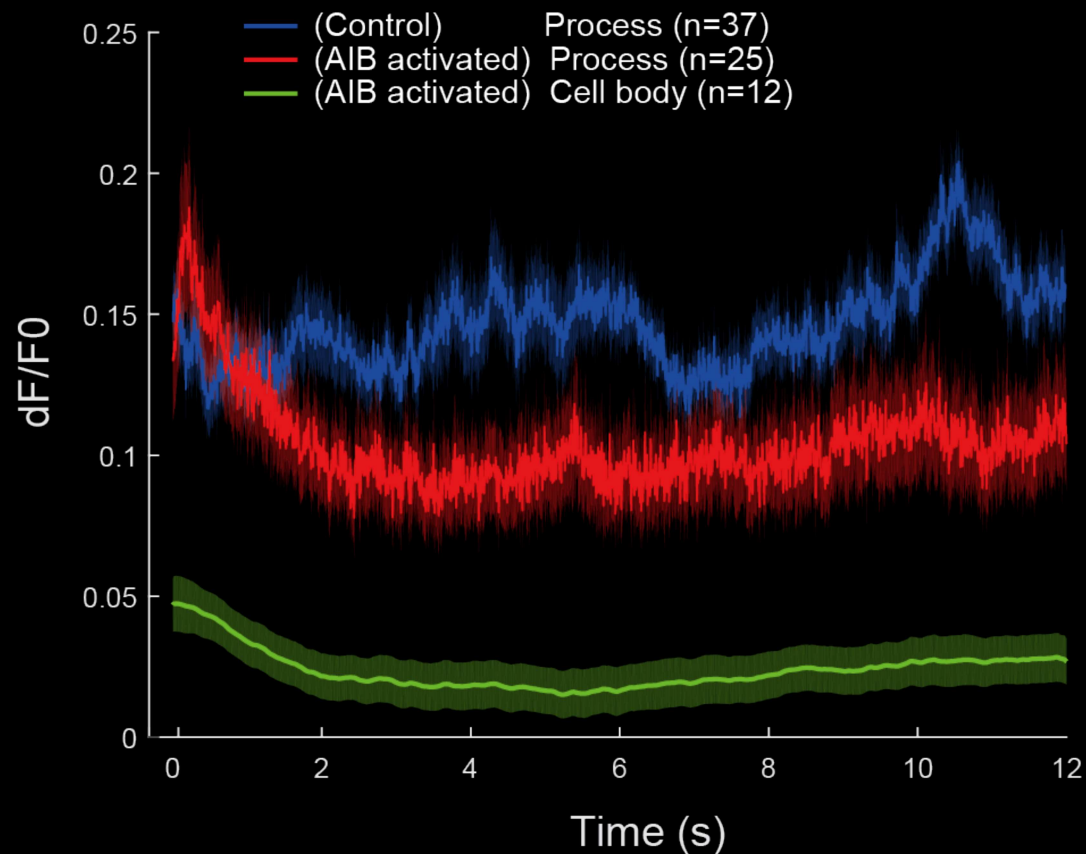


STD allows energy landscape to change over time



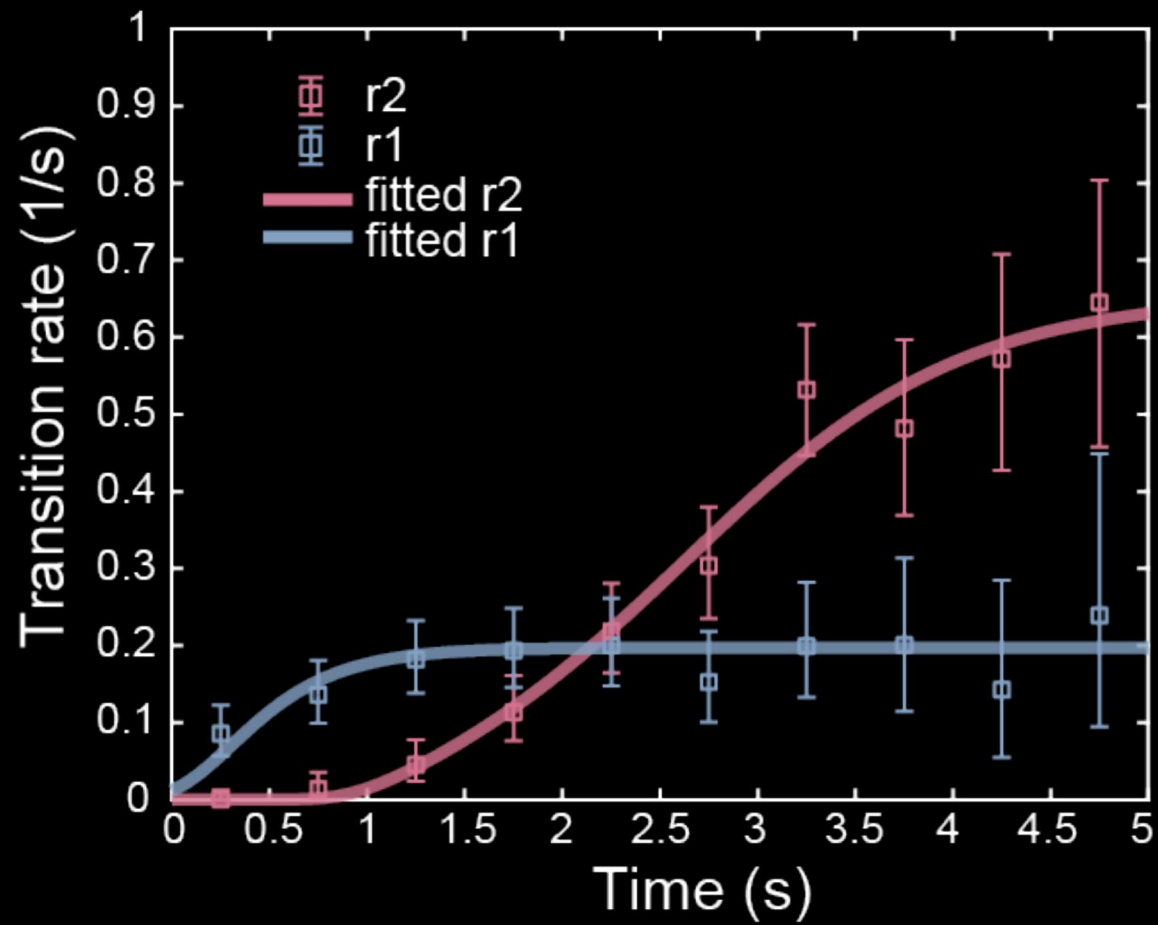
Glutamatergic neurotransmission reveals the time scale of STD

AIB::Chrimson; RIB::iGluSnFR
(immobilized, t=0 means stimulation began)



$$\frac{\Delta F(t)}{F_0} = B(e^{-t/\tau_1} - e^{-t/\tau_2})$$

$$\tau_1 = 0.9 \text{ s}, \tau_2 = 100 \text{ ms}$$

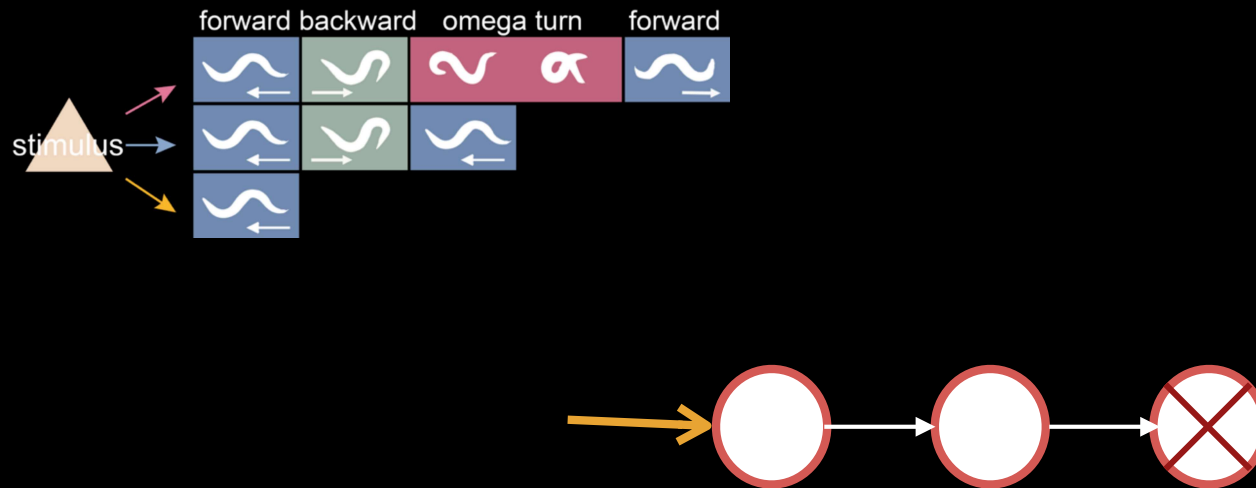


$$r_2(t) = \frac{k}{\pi} \operatorname{erfi}^{-1} \left(\alpha + \beta e^{-t/\tau_g} \right)$$

$$\operatorname{erfi}(x) = \int_0^x e^{z^2} dz$$

$$\tau_g = 0.9s$$

How is the activity in one module terminated?



Abele M, 1991
*Corticonics: Neural Circuit
of the Cerebral Cortex*
Cambridge University Press

Control

ALM/AVM::ChR2

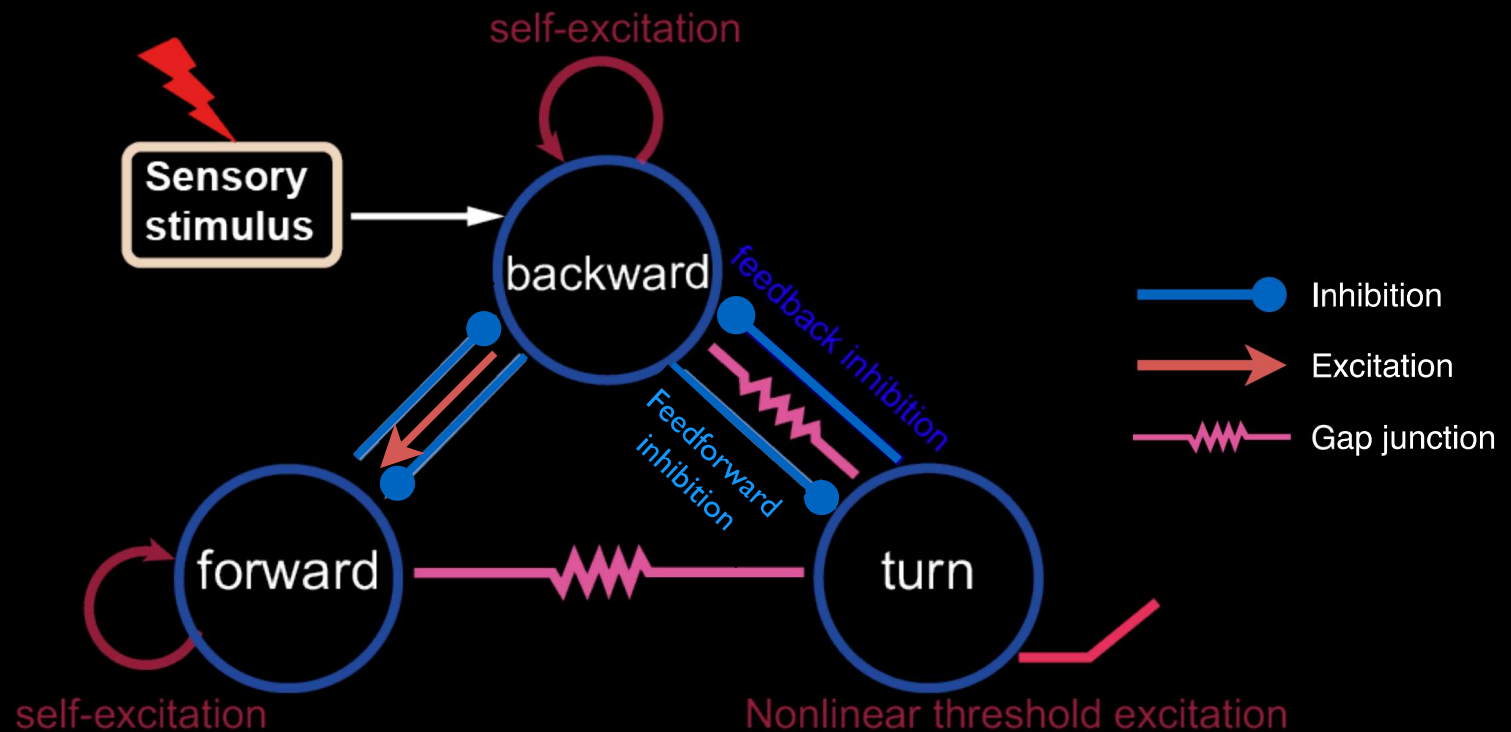
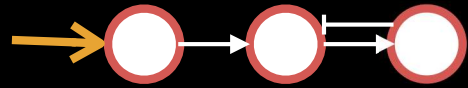
1.5 sec activation

Ablating the turning module

ALM/AVM::ChR2
P_{lin-4}::miniSOG

1.5 sec activation

Feedback Inhibition terminates the backward state



Motor Sequence at a Long Timescale

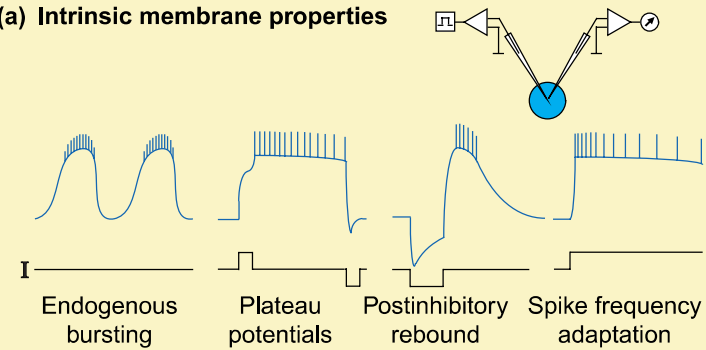
- Feedforward excitatory couplings between motor modules robustly trigger motor sequences.
- A winner-take-all operation with mutual inhibitions between motor modules flexibly controls the timing of a sequence.
- Short term synaptic depression could modulate the probability of motor state transition.

Motor Sequence at a Short Timescale

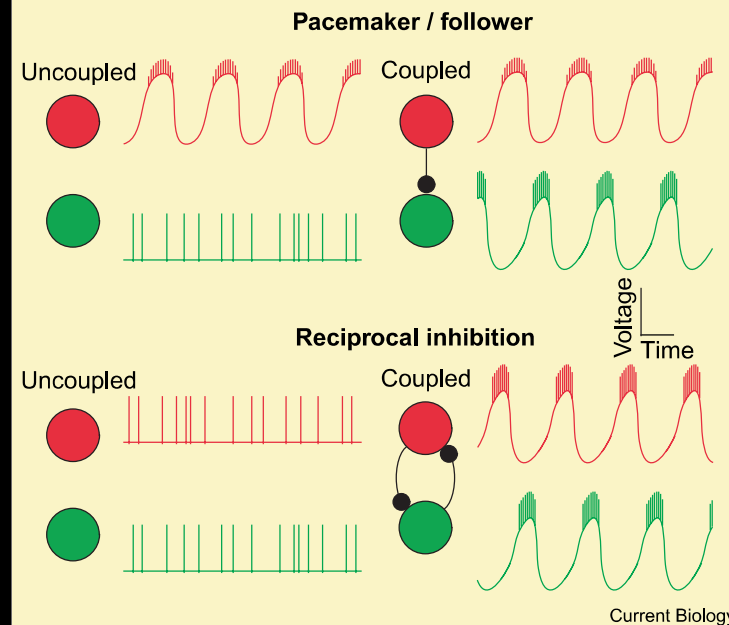
— Rhythmic Motion

Central Pattern Generator

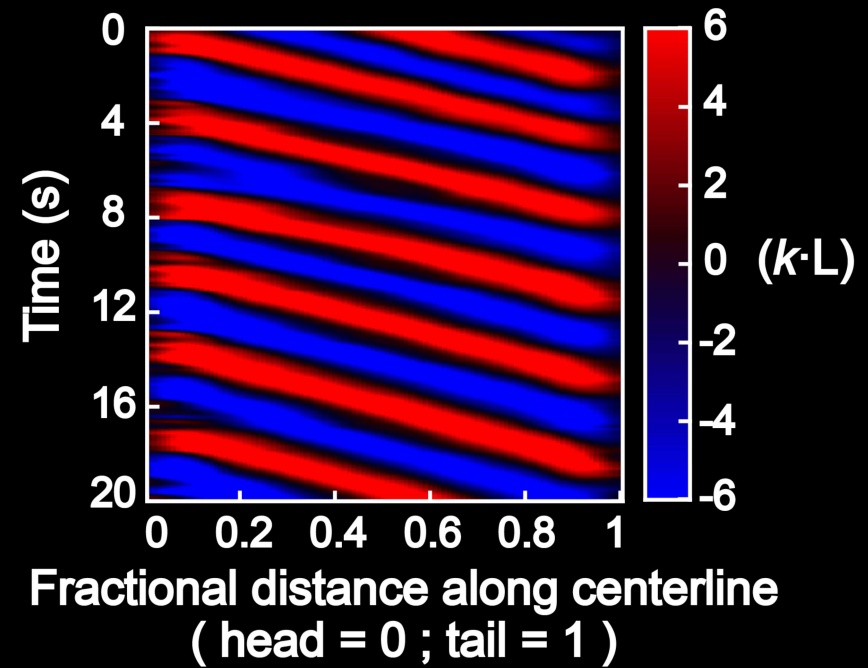
(a) Intrinsic membrane properties



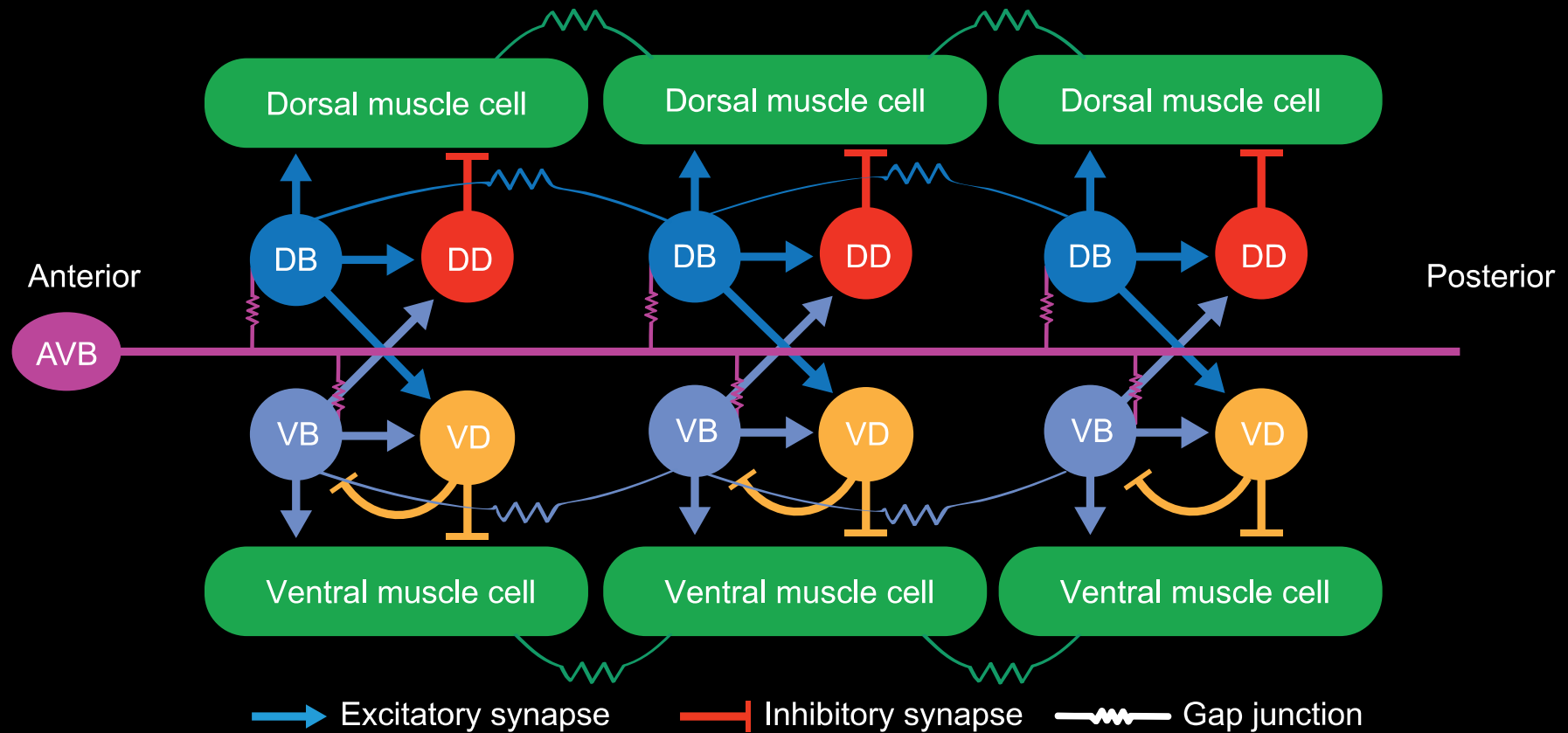
(b) Mechanisms of rhythm generation



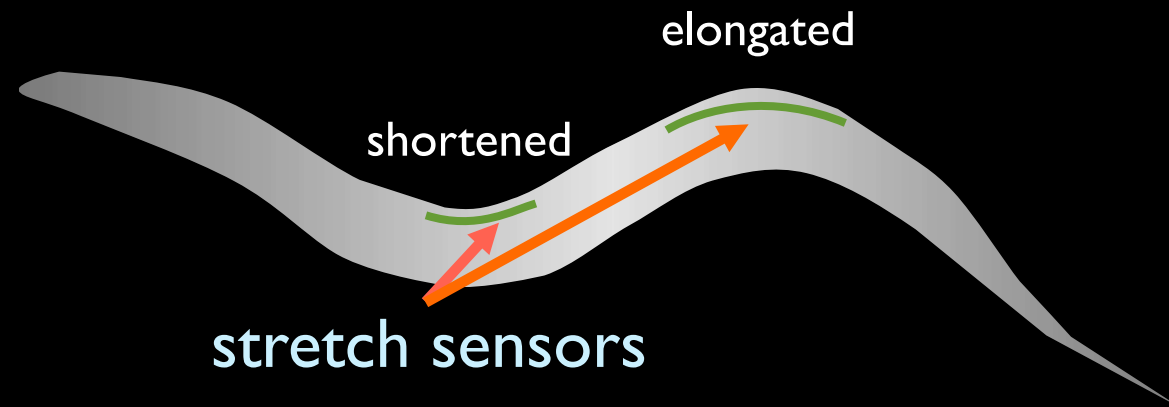
Propagating bending waves across spacetime



The nerve cord that back-propagates the bending waves

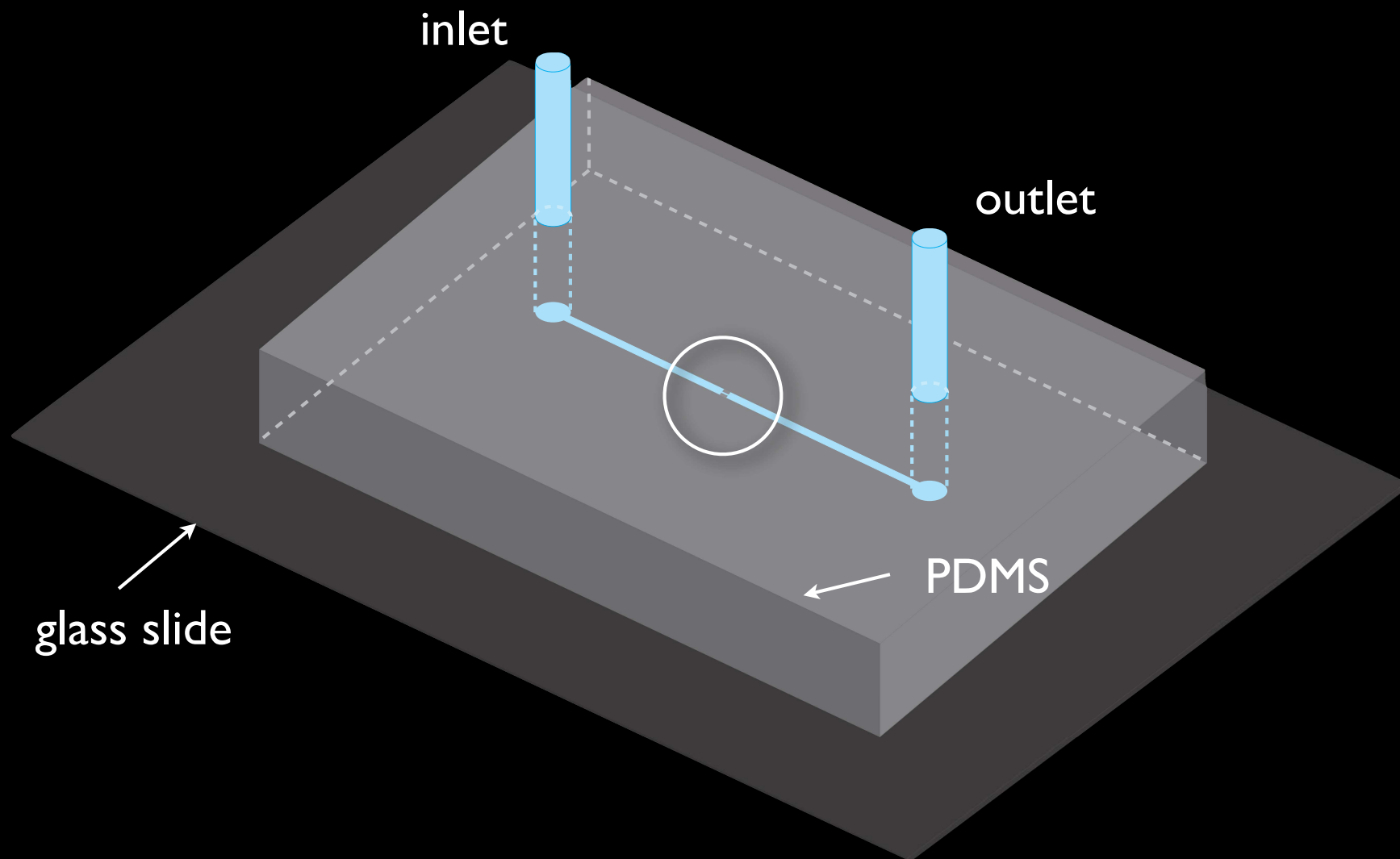


Proprioception

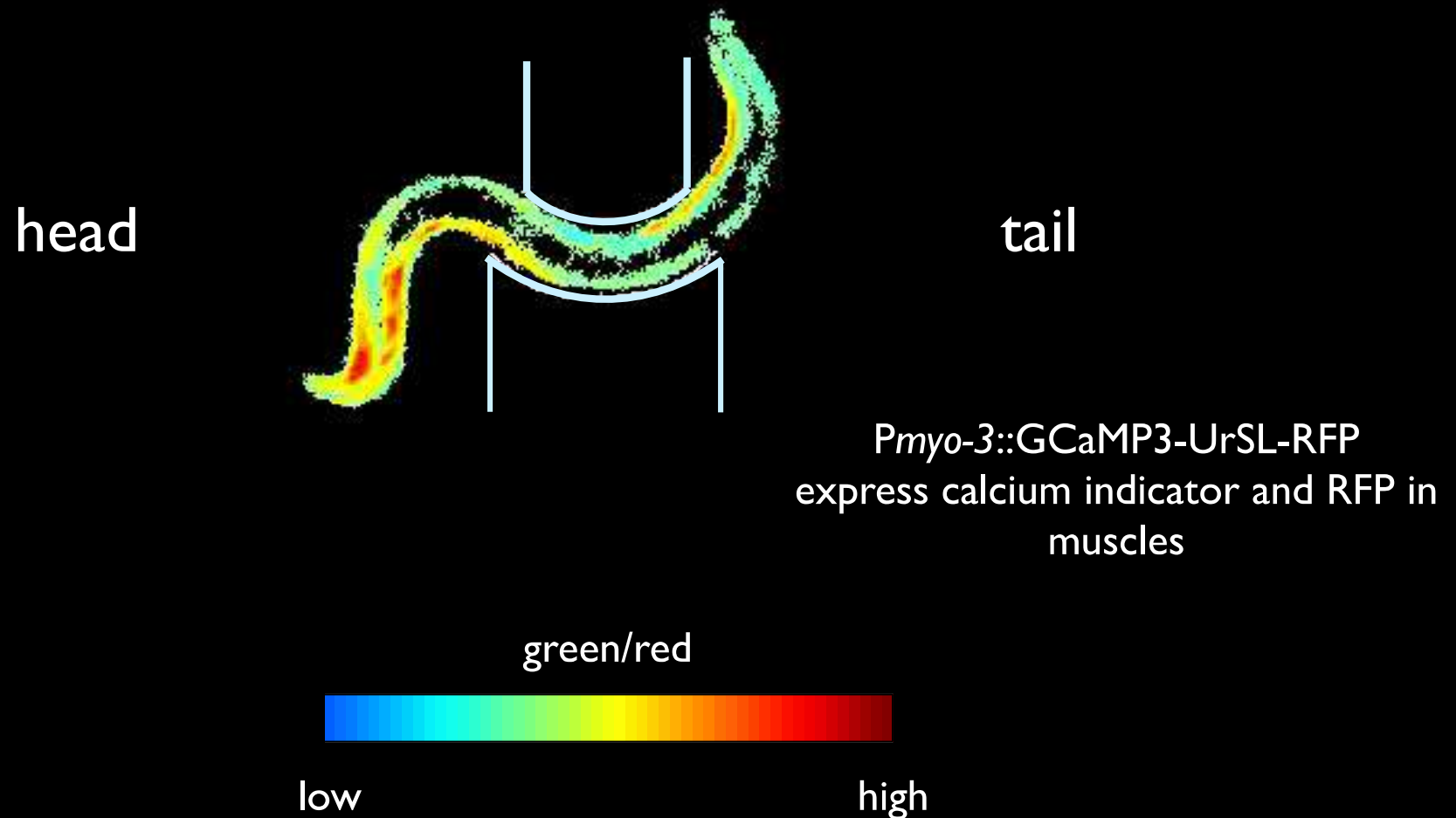


Richard L Russell 1970s
Wen et al, *Neuron* 2012

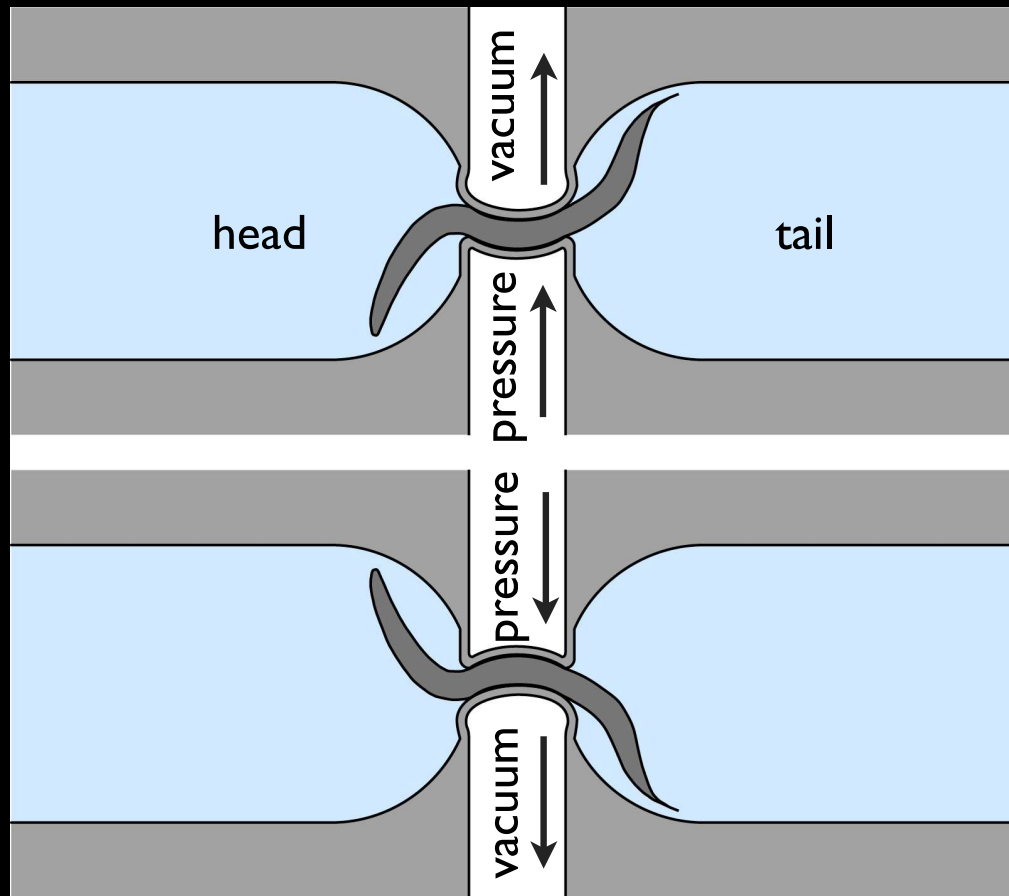
Microfluidics



Posterior body region is compelled to bend in the same direction as its anterior neighbor

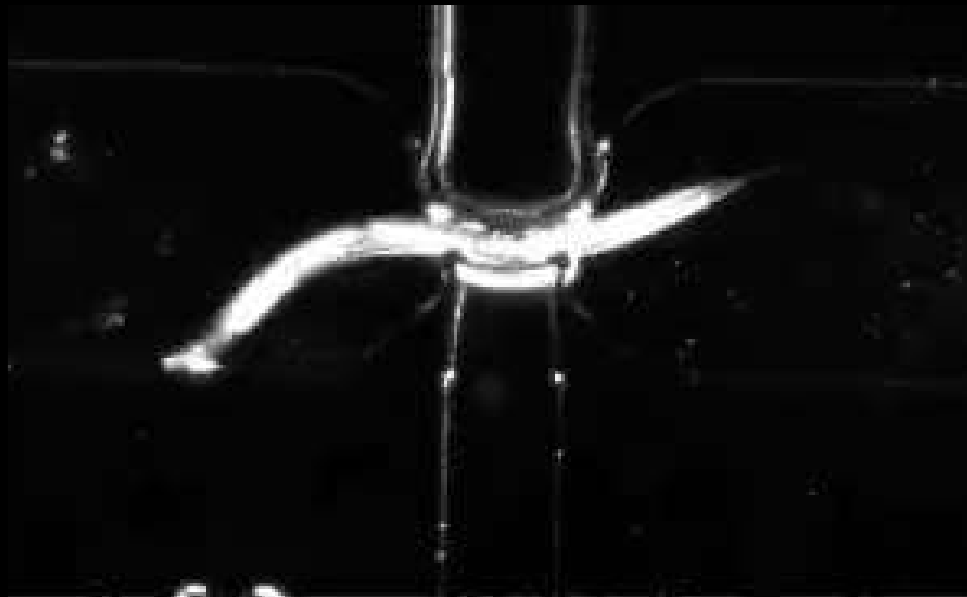


Pneumatic microfluidic device

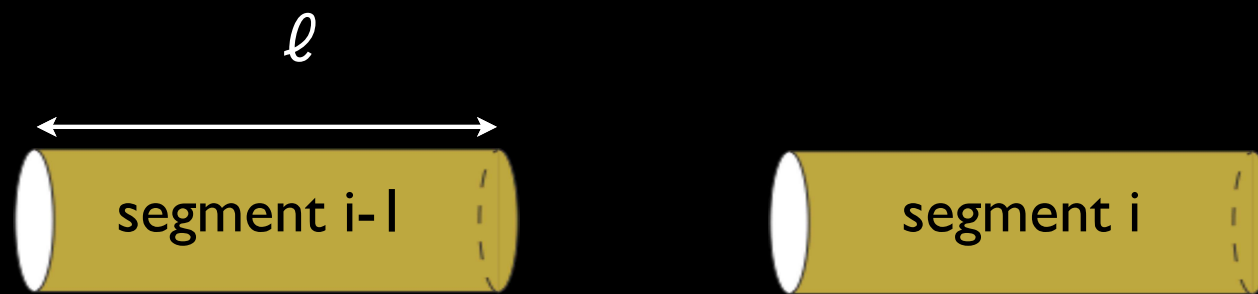


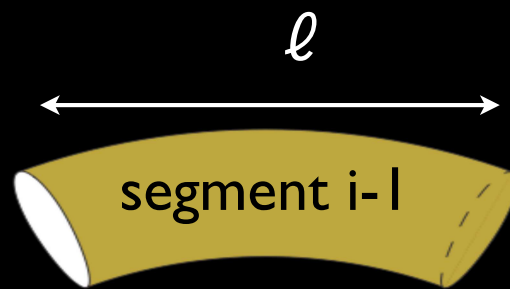
more viscous medium

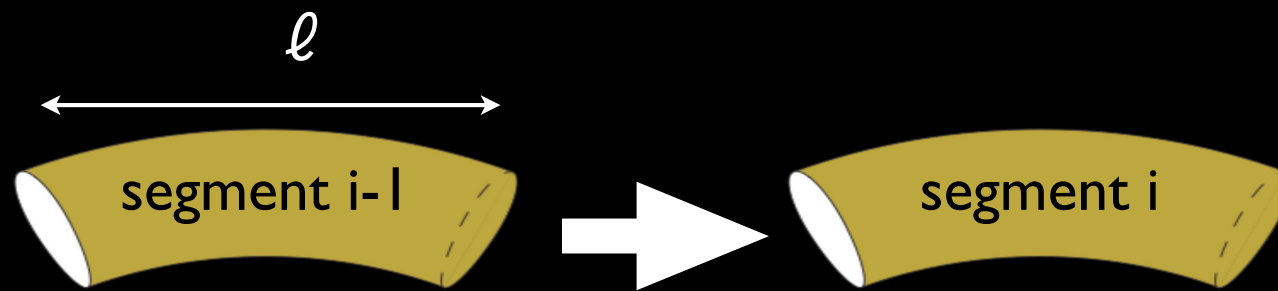
head



tail







A linear proprioceptive coupling model



$$\tau_c \frac{dv_i}{dt} = -v_i + c\kappa_{i-1}$$

$$\tau_\eta \frac{d\kappa_i}{dt} = -\kappa_i + v_i \alpha_{max} / b$$

$$\tau_\eta \sim \eta \lambda^4 / b$$

κ : segment curvature

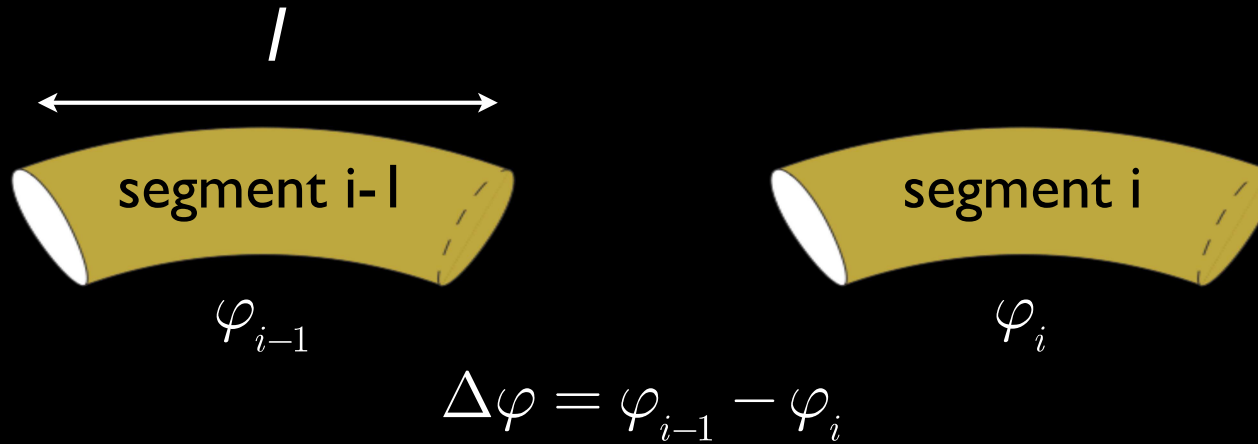
v : relative motor neuron activity

τ_c : internal coupling delay within the circuit
(say cell membrane time constant)

τ_η : delay between motor neuron activities and
local body curvature imposed by external load

η : viscosity of a medium (*given*)

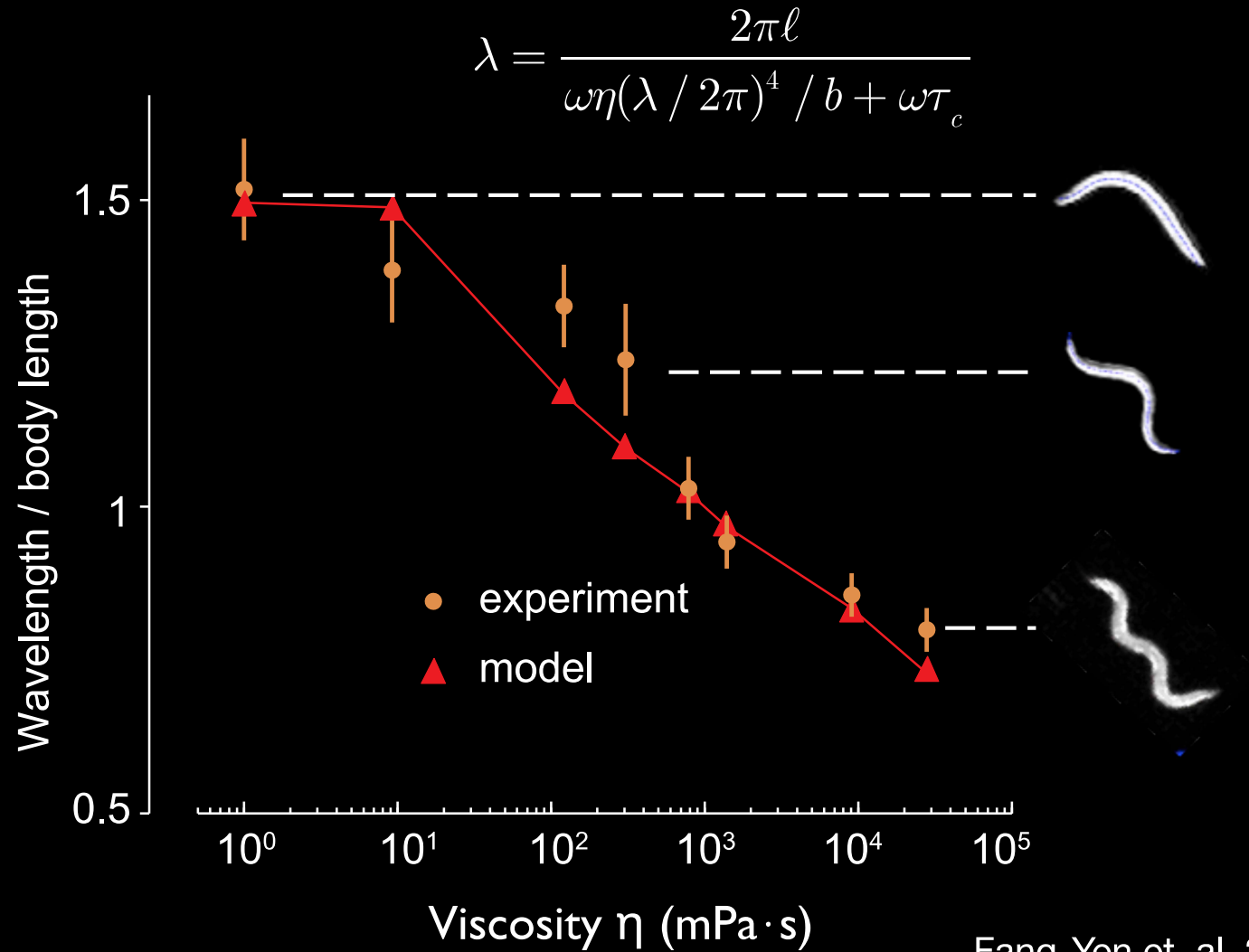
b : bending modulus that reflects
the stiffness of a worm (*measured*)



$$\Delta\varphi \approx \omega(\tau_c + \tau_\eta) \quad \omega: \text{angular undulation frequency (measured)}$$

$$\lambda = \frac{2\pi}{\Delta\varphi} \ell \quad \begin{array}{l} \lambda: \text{undulation wavelength} \\ \ell: \text{spatial length of proprioceptive coupling (measured)} \end{array}$$

Continuous gait adaptation



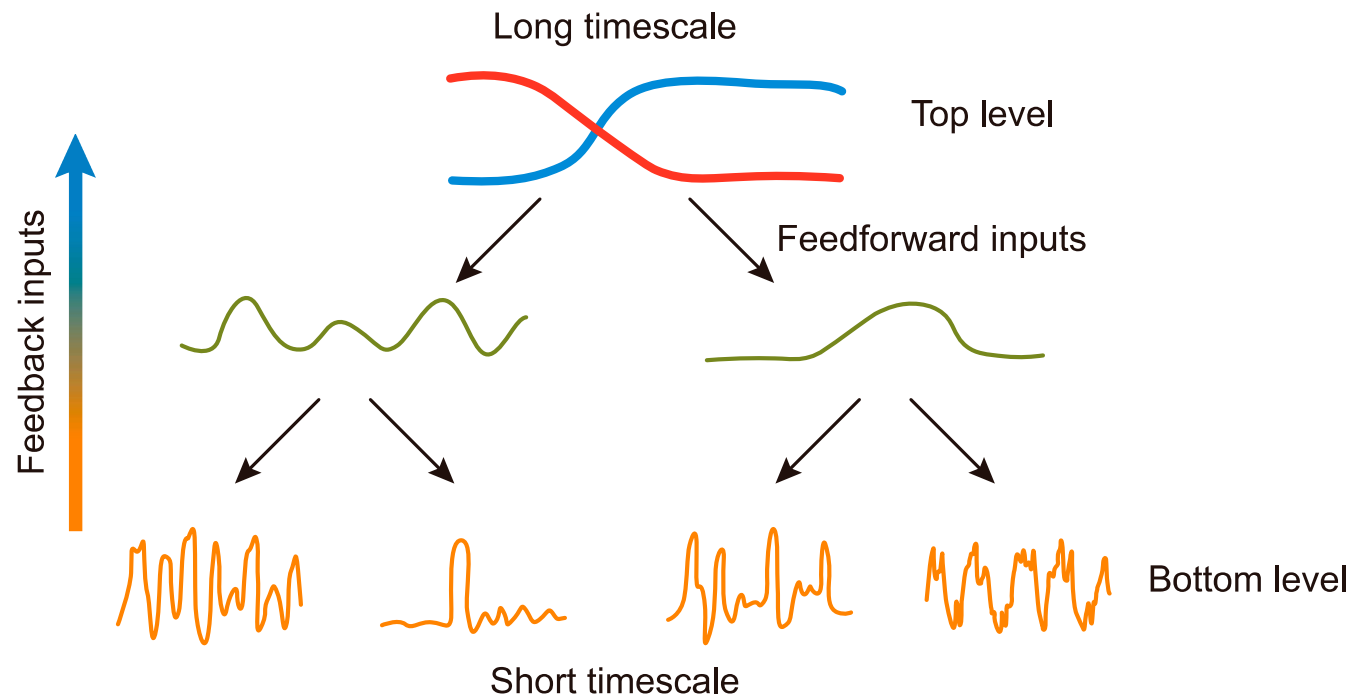
Fang-Yen et. al, 2009

Wen Q. et al, 2012

Motor Sequence at Short Timescale

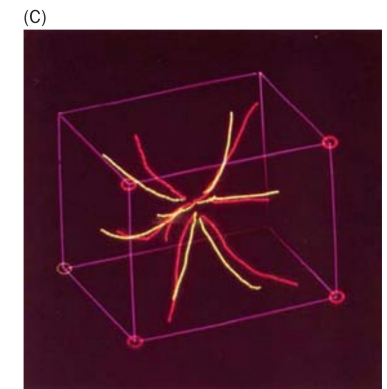
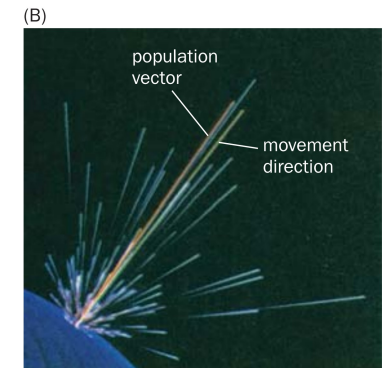
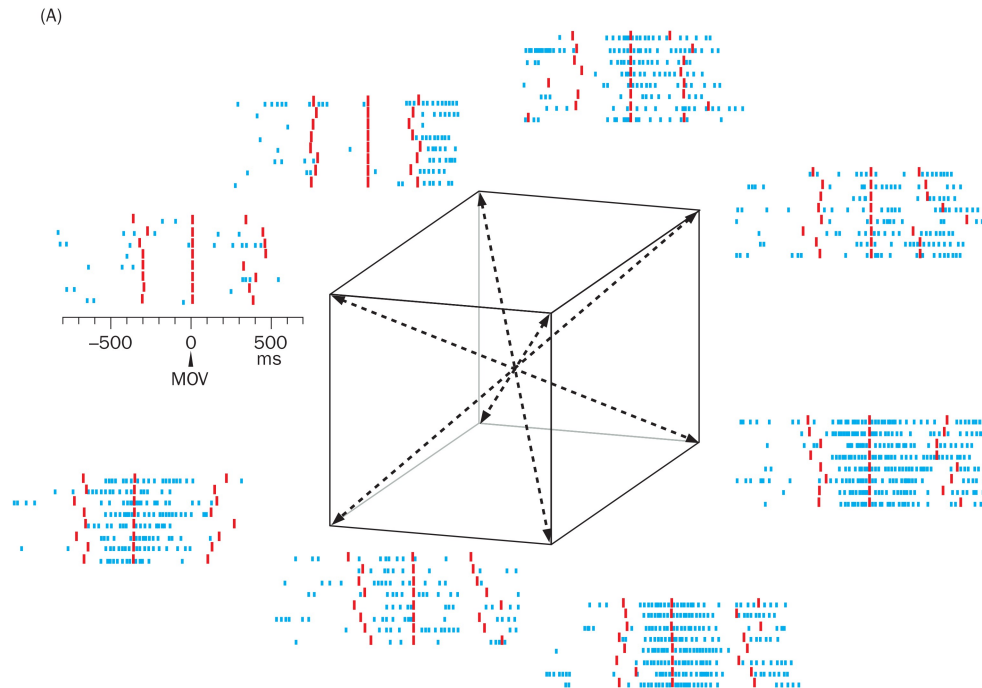
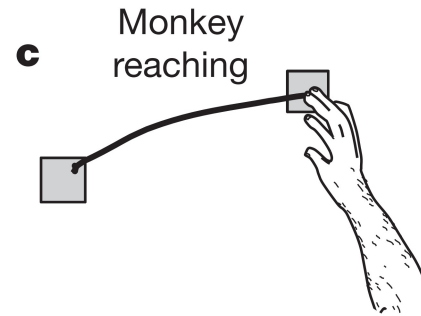
- Motor neurons generate fast rhythms.
- Local proprioceptive coupling propagates the bending activity along the worm body, which can also be viewed as a feedforward chain model.

The hierarchical architecture for movement control



Movements that go beyond rhythmic gait

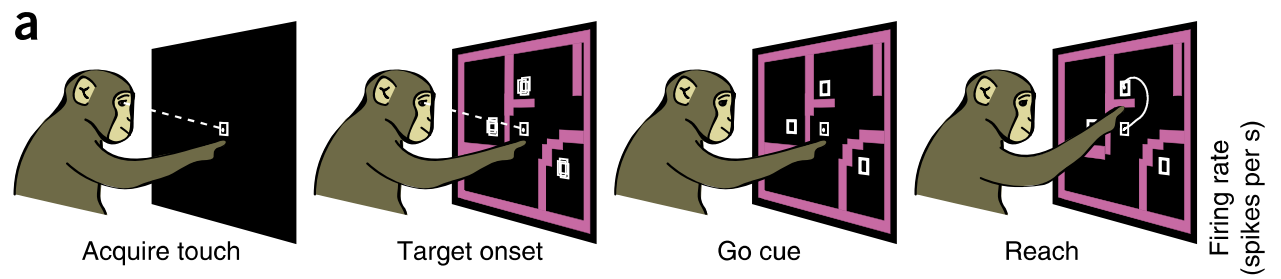
Recordings from motor cortex



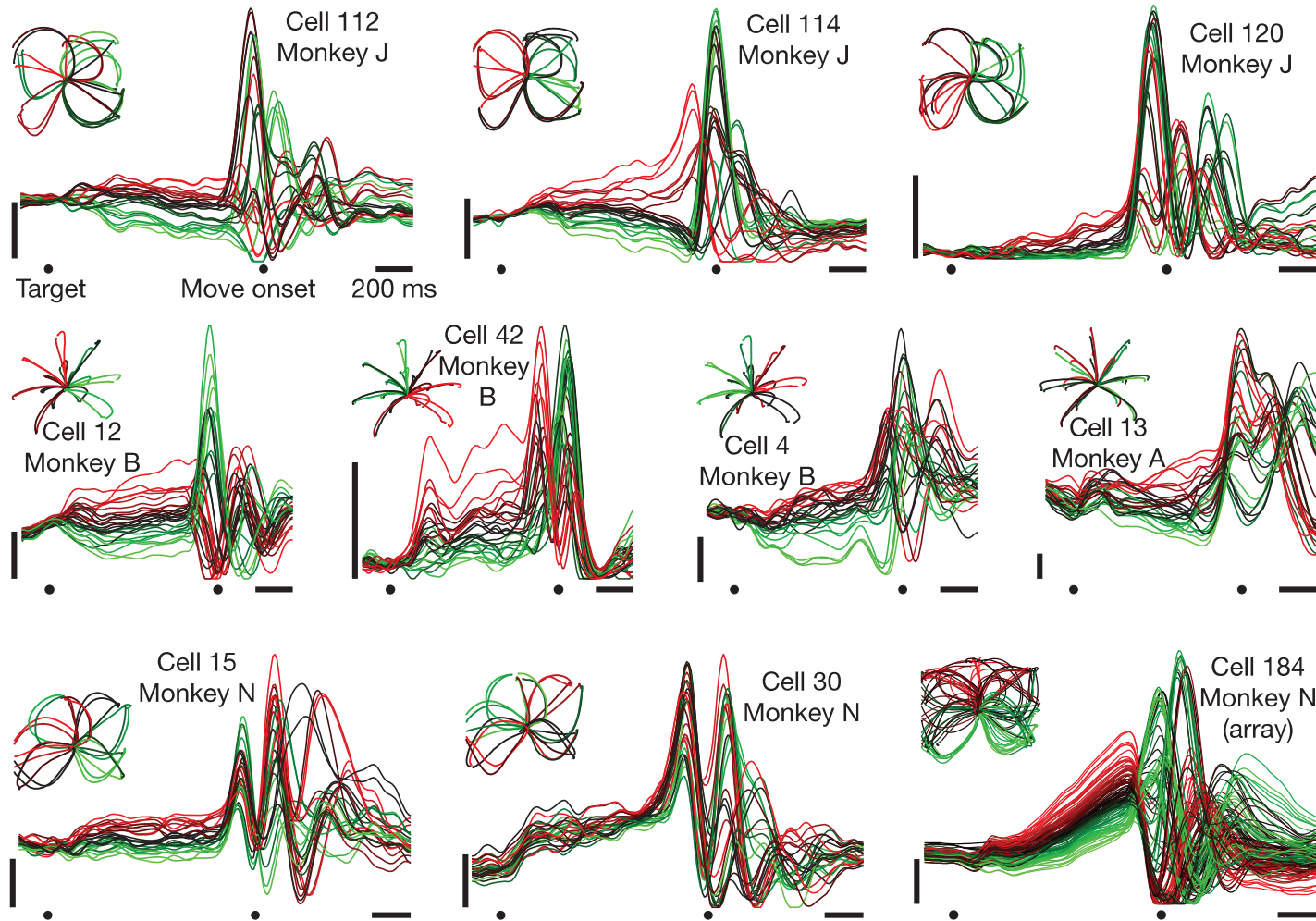
$$r(t) = f(\text{param}_1(t), \text{param}_2(t), \text{param}_3(t), \dots)$$

param = movement direction, ..., movement speed, ..., etc

Monkey reaching task (updated)



Complicated response patterns



From single neuron dynamics to population dynamics

$$\frac{d\vec{r}(t)}{dt} = f(\vec{r}(t)) + \vec{v}(t)$$

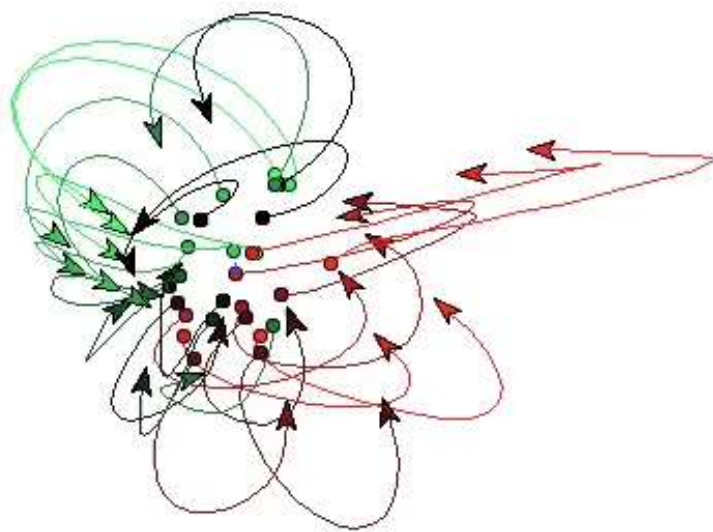


PCA

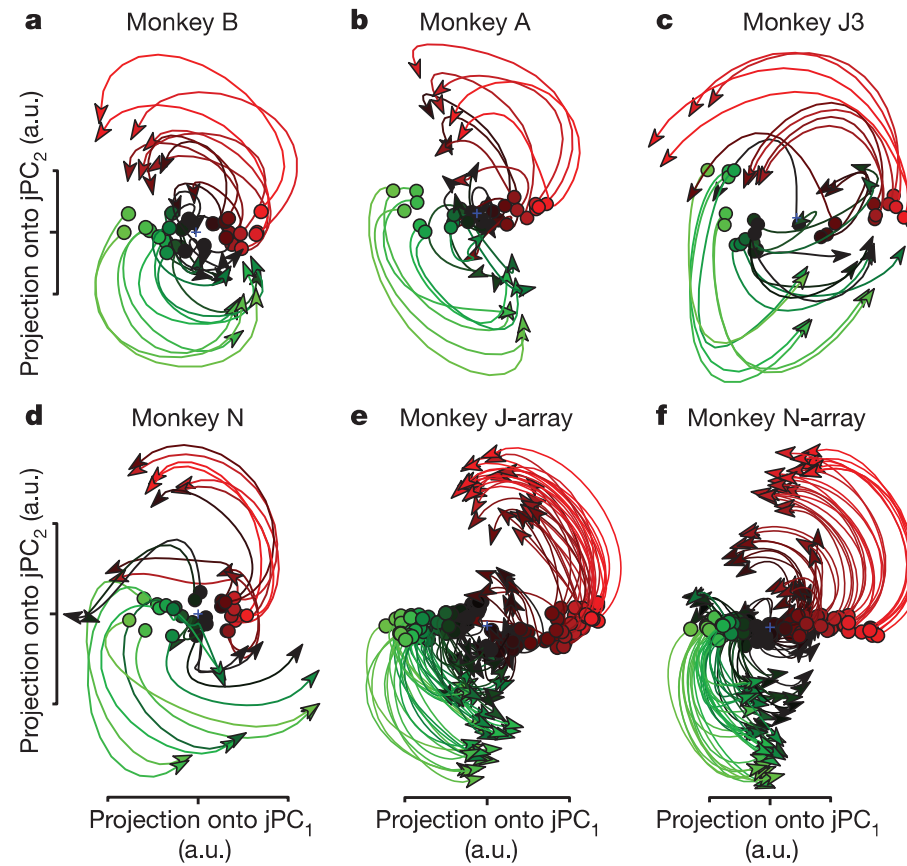
$$\vec{r}(t) = \sum_{i=1}^6 a_i(t) \vec{u}_i$$

$$\frac{d\vec{a}(t)}{dt} = M\vec{a}(t)$$

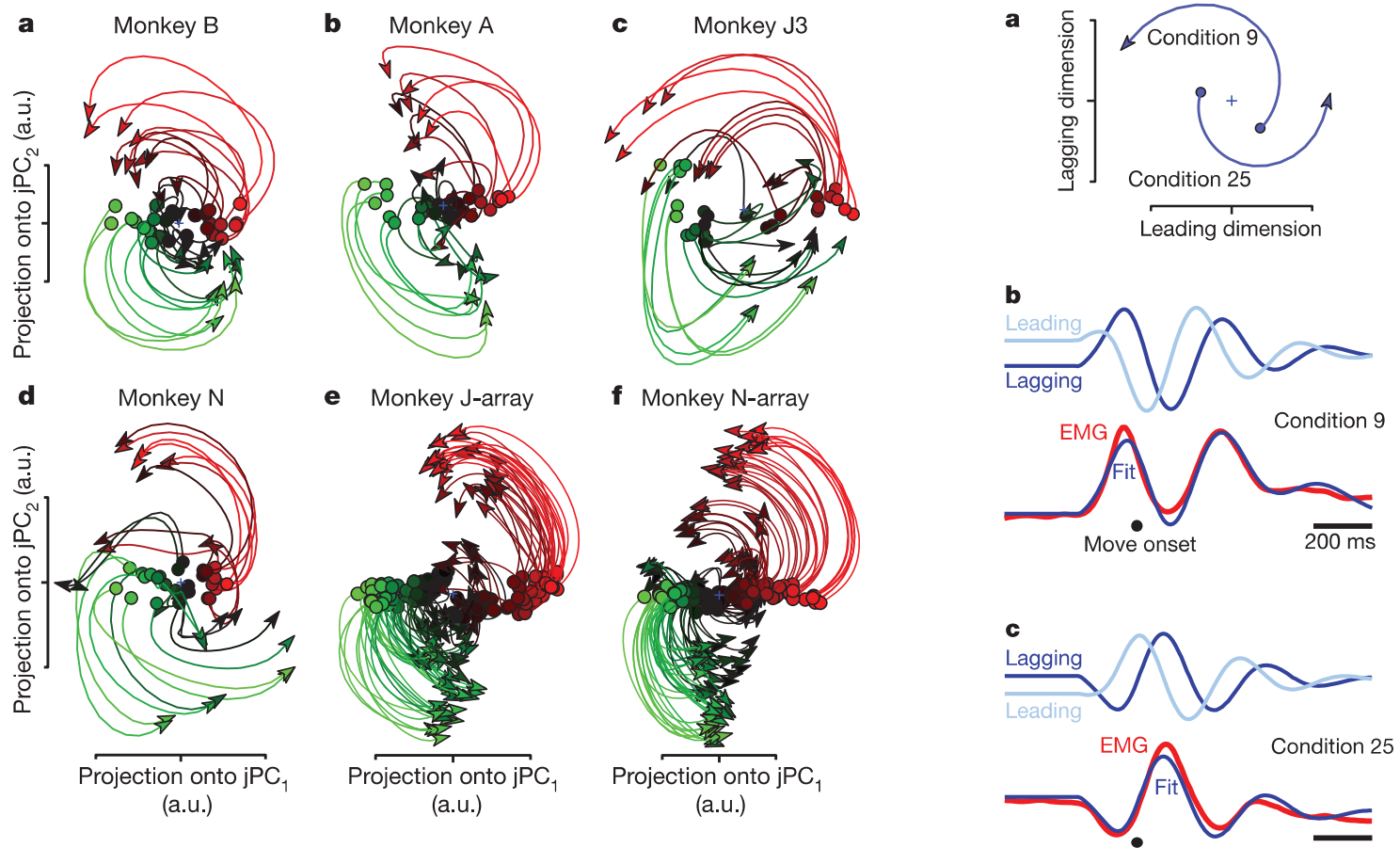
Linear approximation



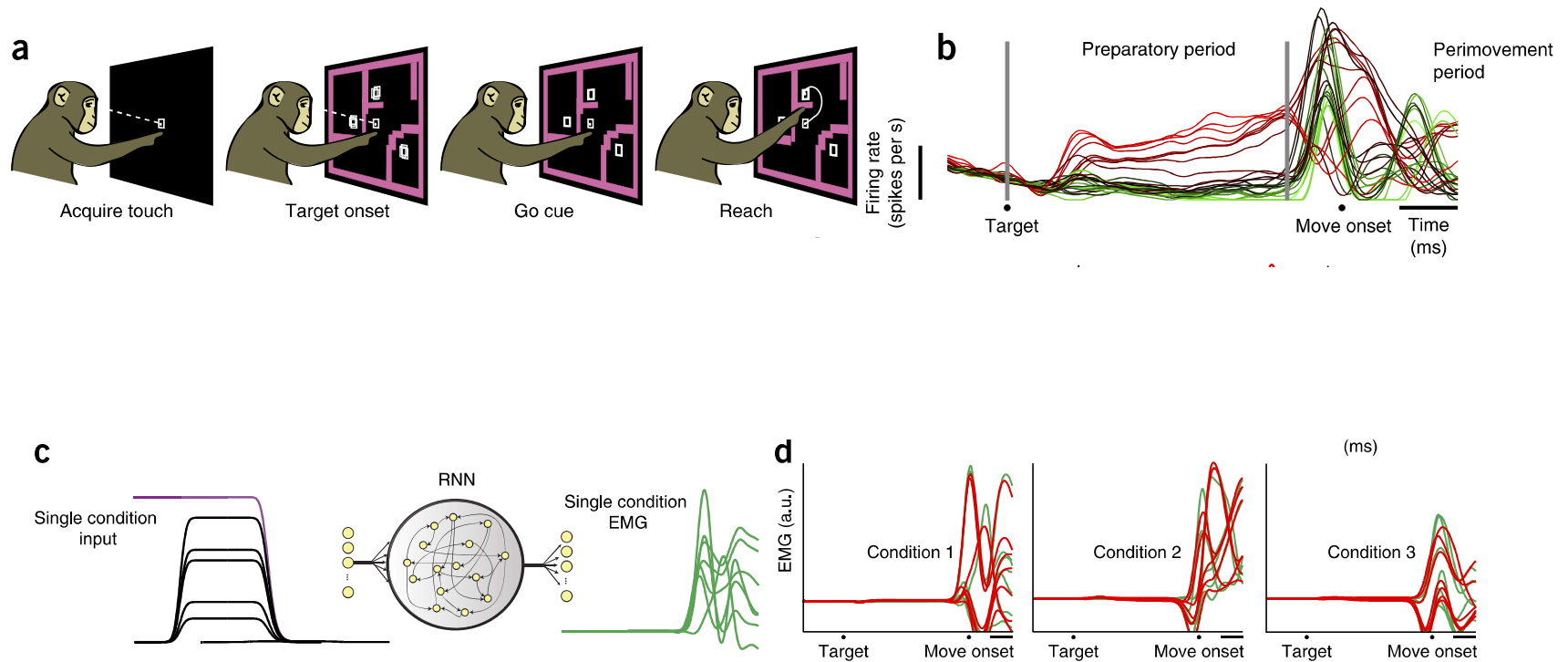
Rotation dynamics during reaching task



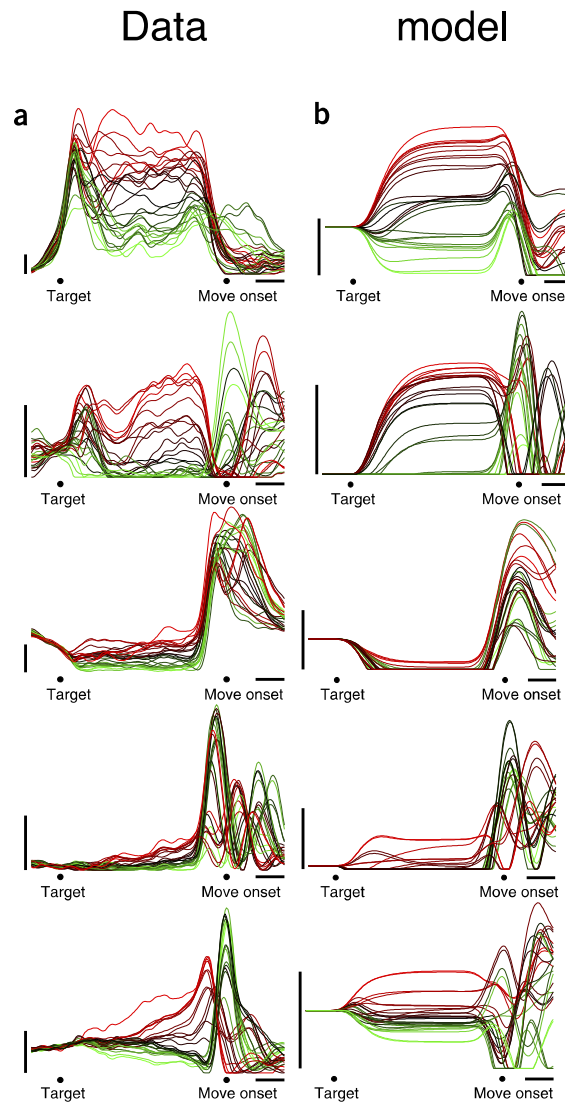
Rotation dynamics during reaching task



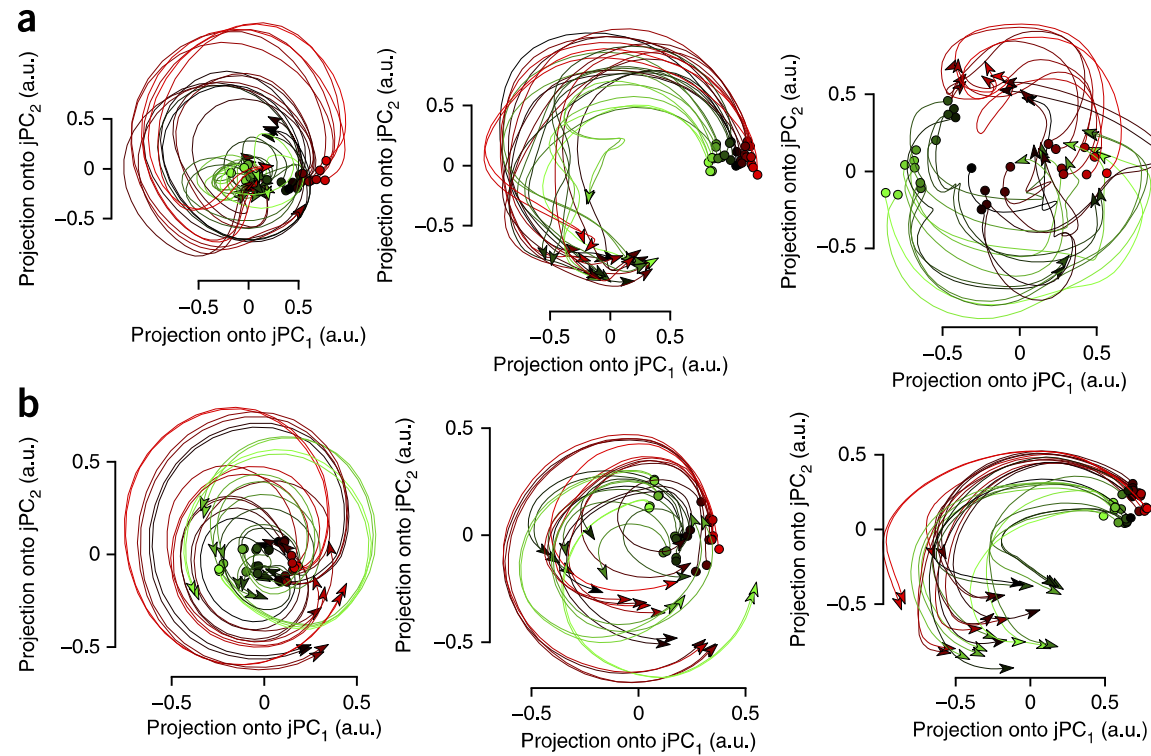
Generate coherent motor patterns from chaotic neural network



Generate coherent motor patterns from chaotic neural network



RNN model can reproduce the rotation dynamics on the population level



What does rotational dynamics really mean?